NEURAL NETWORKS / DAY 3

Energy modelling capstone

A reproducible baseline, a neural candidate, and a defensible conclusion

Full day · 09:00–17:00 | 3 supplied projects | 4 milestone checkpoints | 5-minute team presentations

Today's plan

09:00	Welcome and the brief	Lecture
09:35	ESP gas lock briefing	Lecture
09:55	PE log curve briefing	Lecture
10:15	Break	Break
10:30	Sensor briefing	Lecture
10:45	Sprint 1: data contract	Project work
11:20	Checkpoint 1	Check
11:30	Tabular-NN playbook	Lecture
11:50	Sprint 2: baselines	Project work
12:30	Checkpoint 2	Check
12:40	Lunch	Lunch
13:25	Experiment design	Lecture
13:40	Sprint 3: experiment	Project work
14:20	Checkpoint 3	Check
14:30	Error analysis	Lecture
14:45	Sprint 4: final test	Project work
15:25	Checkpoint 4	Check
15:35	Break	Break
15:45	Presenting evidence	Lecture
16:00	Sprint 5: handoff	Project work
16:15	Final presentations	Presentation
16:50	Takeaways, exit check	Discussion

LEARNING OUTCOMES

What you will be able to do

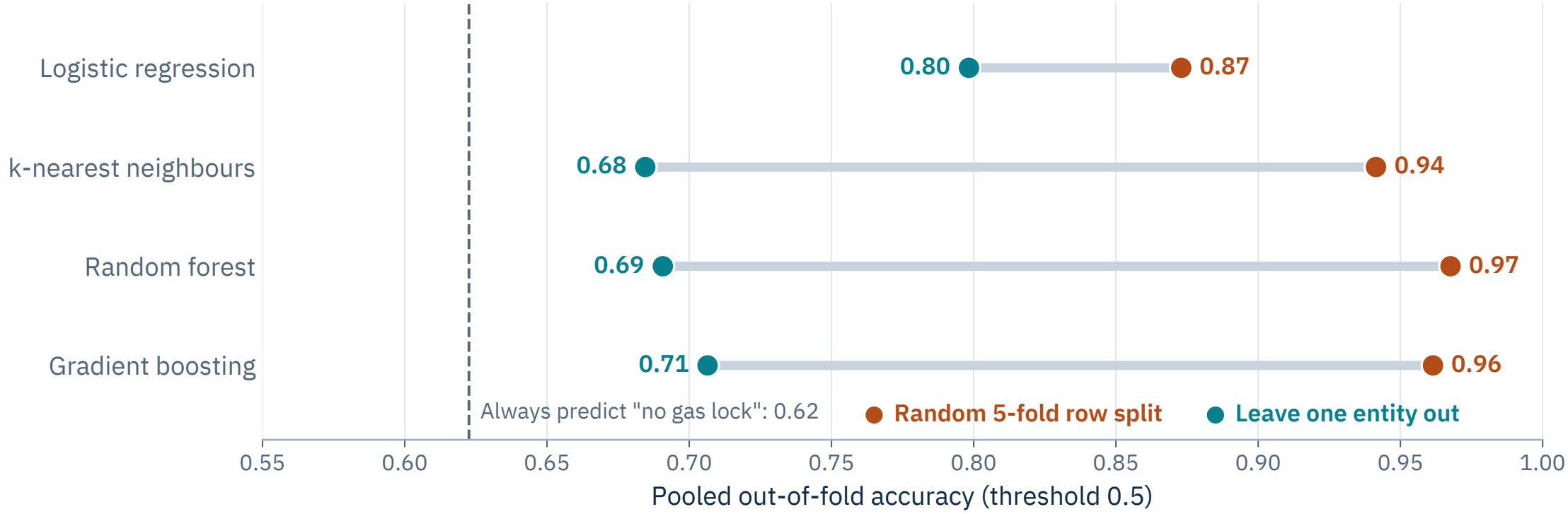
By 17:00 you will be able to:

1. Write a data contract and an audit for a supplied energy dataset.
2. Choose a validation design that matches what the data can establish.
3. Compare a baseline and a small neural network on identical rows and preprocessing.
4. Run one controlled experiment, then lock the configuration before the final test.
5. Diagnose errors by group, class and calibration, and state their limits.
6. Present the evidence in five minutes with a run that others can reproduce.

MOTIVATING CASE

Random row splits flattered every ESP model

Same file, same four models, two validation designs. Which number answers "will it work on the next pump"?



Computed from esp_train.csv (3,243 rows, 7 entities): pooled out-of-fold accuracy at threshold 0.5, preprocessing fitted inside each fold.

PART 1 OF 7

01

The capstone brief

Sound evidence, not a target score

- The deliverable and the project lifecycle
- The day plan and four milestone checkpoints
- Three roles for data and a one-page data contract
- An audit checklist applied to all three files

The deliverable is evidence, not a winning model

Define

- The target and the row unit
- What the evaluation can establish

Compare

- A simple baseline first
- One small network on the same split

Reproduce

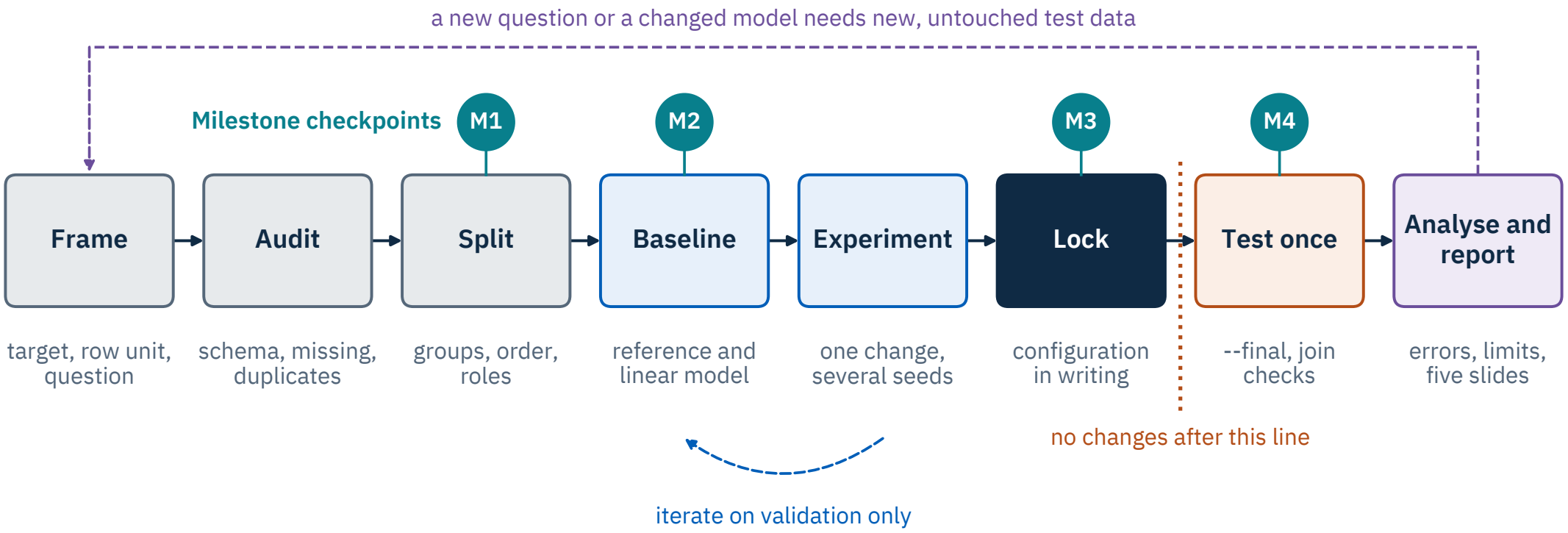
- One command from a fresh process
- Score table and saved artifacts

Present

- Five minutes, five slides
- Limitations stated explicitly

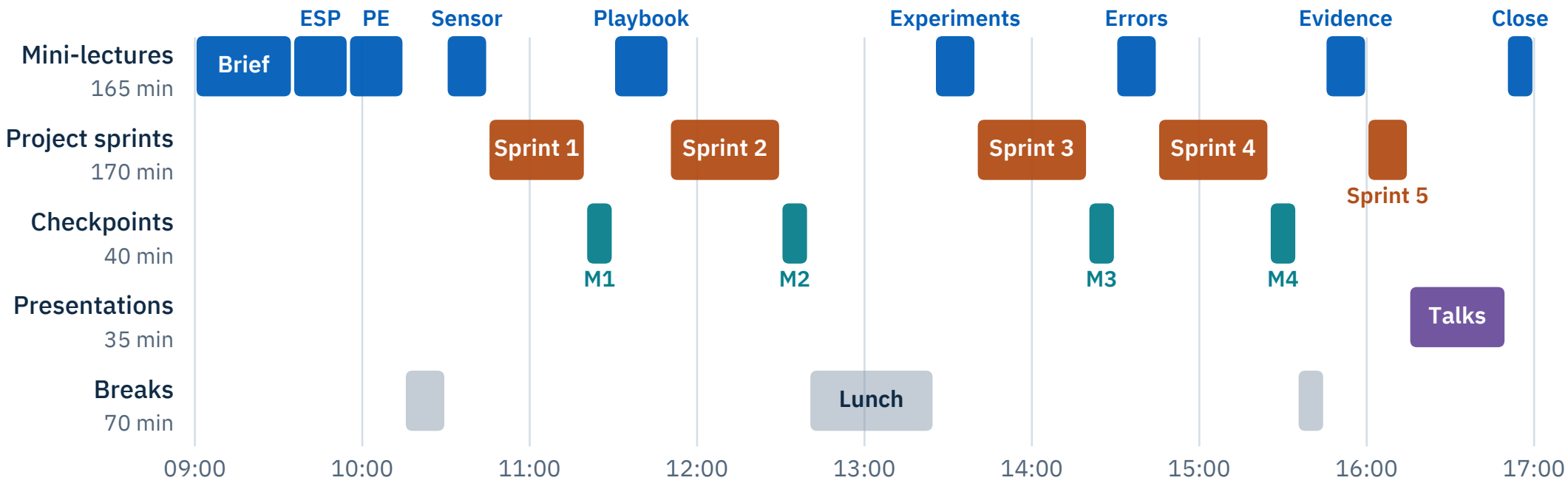
◆ A failed or inconclusive neural model can earn full marks with sound evidence.

Every project follows the same eight-stage lifecycle



Schematic. Teal badges mark the four milestone checkpoints; after the lock only analysis and reporting change.

Short lectures alternate with five project sprints



Schematic of today's plan: 165 min of lectures including the close, 170 min of sprints, four 10-min checkpoints, 35 min of presentations and 70 min of breaks.

Each team chooses one of three supplied projects

Project	Actual target	Supplied training data	Validation fallback
ESP gas lock	GAS_LOCK, values 0 and 1	3,243 rows, 7 entities	Hold out 2 entities
PE log curve	PE, numeric	2,922 rows, 6 input columns	Last 20% of source rows
Sensor signals	target, values −1 and 1	3,580 rows, 10 inputs plus uuid	Stratified 20% of rows

Counts describe the supplied course files.

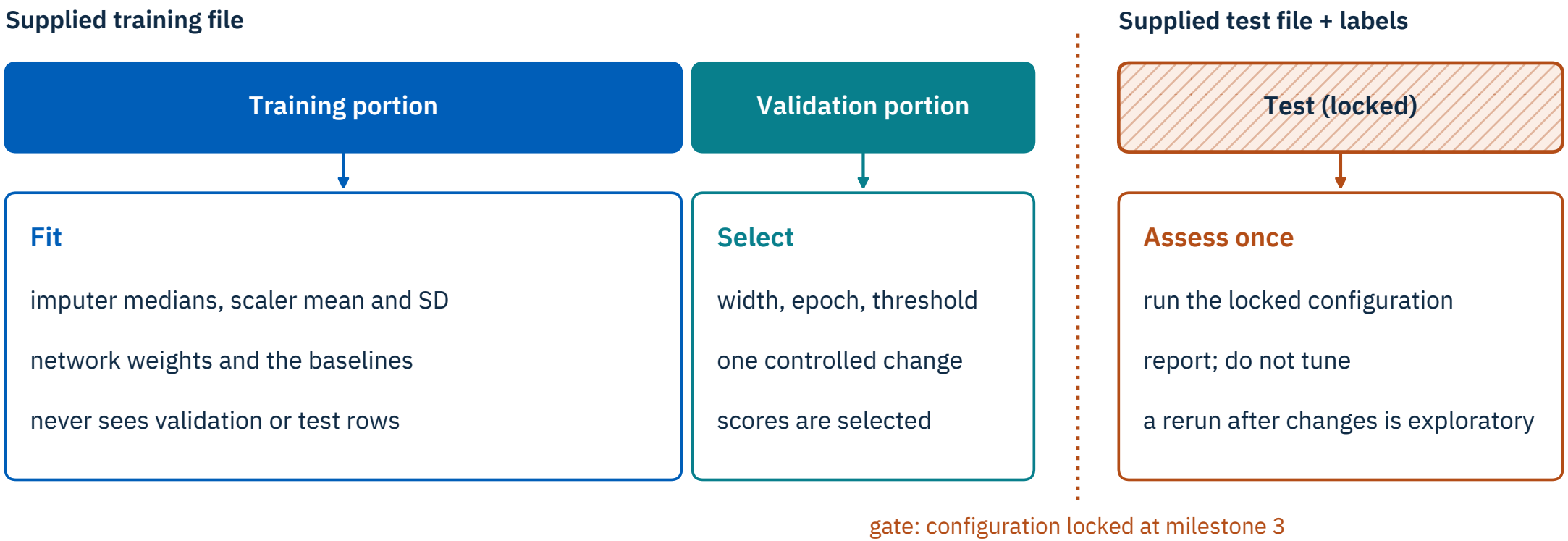
◆ Each fallback split has a stated limit; a team may propose a better split at checkpoint 1.

Four milestones gate the day

Milestone	Due	Evidence to show
M1 Data contract	11:20	Schema, target, audit findings, split proposal
M2 Baseline	12:30	Reproducible validation result and learning curve
M3 Controlled experiment	14:20	Hypothesis, seed spread, locked configuration
M4 Locked final assessment	15:25	Final scores, join checks, one error plot
Presentations	16:15	Five slides, one command, predictions with ids

◆ A team that misses a gate keeps working on it; the gate moves, the evaluation standard does not.

Each portion of the data has one job



Schematic. Validation is carved from the supplied training file; the supplied test file and its labels are opened once, after the configuration is locked at milestone 3.

A data contract fits on one page

Row unit and target

- What one row represents
- Target column and its values

Predictors

- Columns available at prediction time
- Units, or "undocumented"

Excluded metadata

- IDs, keys and timestamps kept out
- The reason for each exclusion

Groups and order

- Entity, depth or record structure
- The split that respects it

Quality

- Missing, duplicate and label counts
- A decision for each finding

Limits

- What the final score cannot show
- Metadata that would remove the limit

◆ Define the split before fitting any preprocessing.

The audit checklist flags issues in every file

	ESP gas lock	PE log curve	Sensor signals
Shape and schema	✓ 3,243 rows × 9 columns	✓ 2,922 rows × 7 columns	✓ 3,580 rows × 12 columns
Target and balance	✓ GAS_LOCK = 1 in 37.7% of rows	✓ PE 0.20–8.09, median 3.51	! labels −1/1 (57.6% = 1): map to 0/1
Missing values	! 300 cells, all in 50 esp_39 rows	✓ none	✓ none
Duplicates	! 2 identical rows; 6 repeat stamps	! 15 rows; a 12-row block repeats	✗ 2,484 rows repeat a feature vector
Keys and joins	✓ uuid unique in test; join on uuid	! no key: labels align by position	✓ uuid unique; labels join on uuid
Groups and order	✓ 7 entities, hourly TimeSlot	✗ no well ID; 7 depth resets	! no groups; uuid order has long runs
Units and meaning	! units undocumented	! units undocumented	! meaning and units undocumented
Test file inputs	✓ 3 unseen entities, same dates	✓ 310 rows, Depth 2964.5–3122.5	✗ 646 of 895 repeat a training row

Computed from the supplied training files and the test input files; no test labels were read. Tick: fine. Gold: needs a documented decision. Orange cross: limits what the evaluation can show.

PRACTICE

4 min · Pairs

Which data may each step use?

For each step, answer: training portion, validation portion, test file, or none.

1. Computing the medians used for imputation
2. Choosing the epoch at which training stops
3. Picking the decision threshold for gas-lock alerts
4. Confirming that test uuids match the label file one-to-one
5. Choosing width 32 over 128 after seeing both test scores

PART 2 OF 7

02

Project 1: ESP gas lock

Classifying labelled operating observations on held-out entities

- The actual schema and label prevalence
- Traces, class distributions and missing rows
- The validation question and a grouped split
- Baseline, candidate and the final join

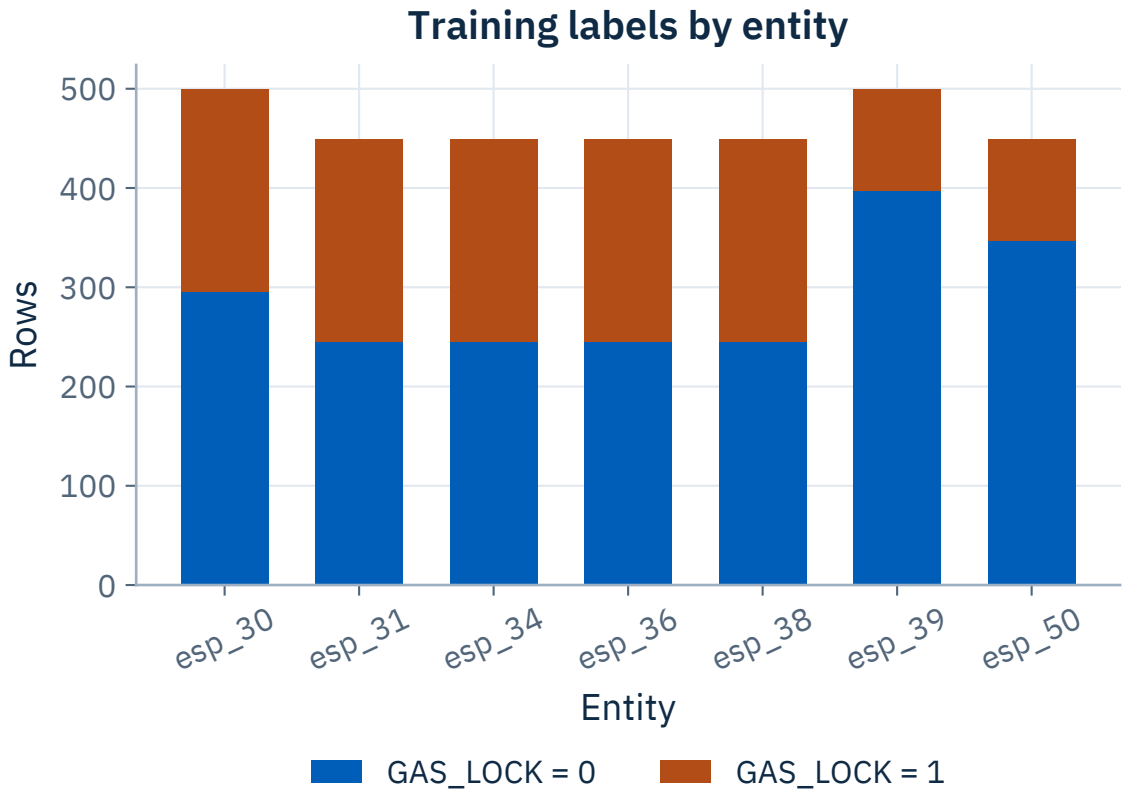
Six sensors, an entity and a timestamp per row

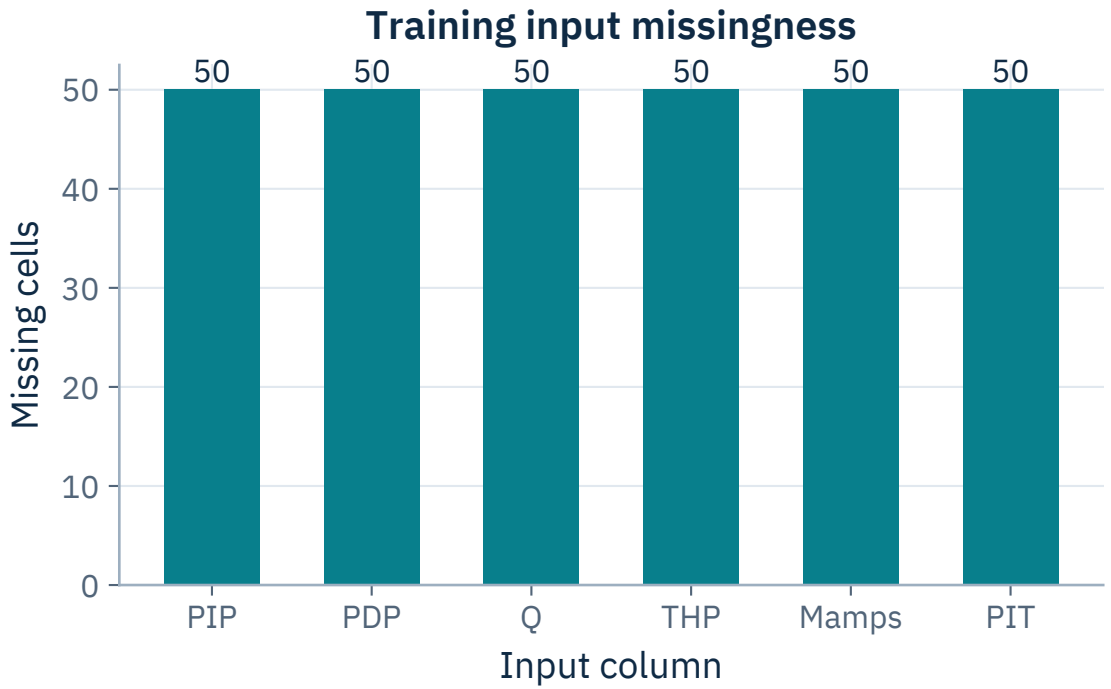
Role	Columns	Treatment
Numeric measurements	PIP, PDP, Q, THP, Mamps, PIT	Check missingness and units
Asset grouping	ENTITY	Hold entities together in validation
Ordering	TimeSlot	Parse timestamps, inspect chronology
Target / test key	GAS_LOCK / uuid	Target in train, join key in test

No vibration channels are present in the supplied files.

◆ ENTITY and TimeSlot define the split and the order; they are not model inputs.

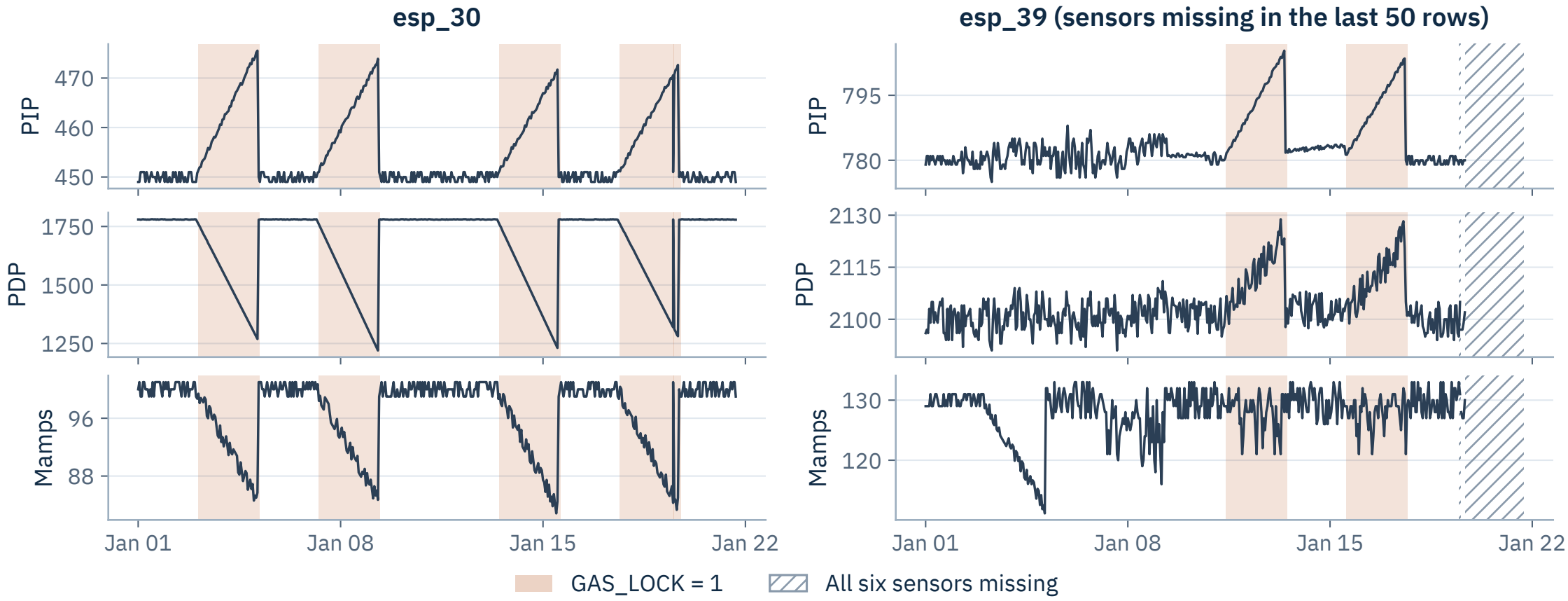
Label prevalence is measured, not assumed





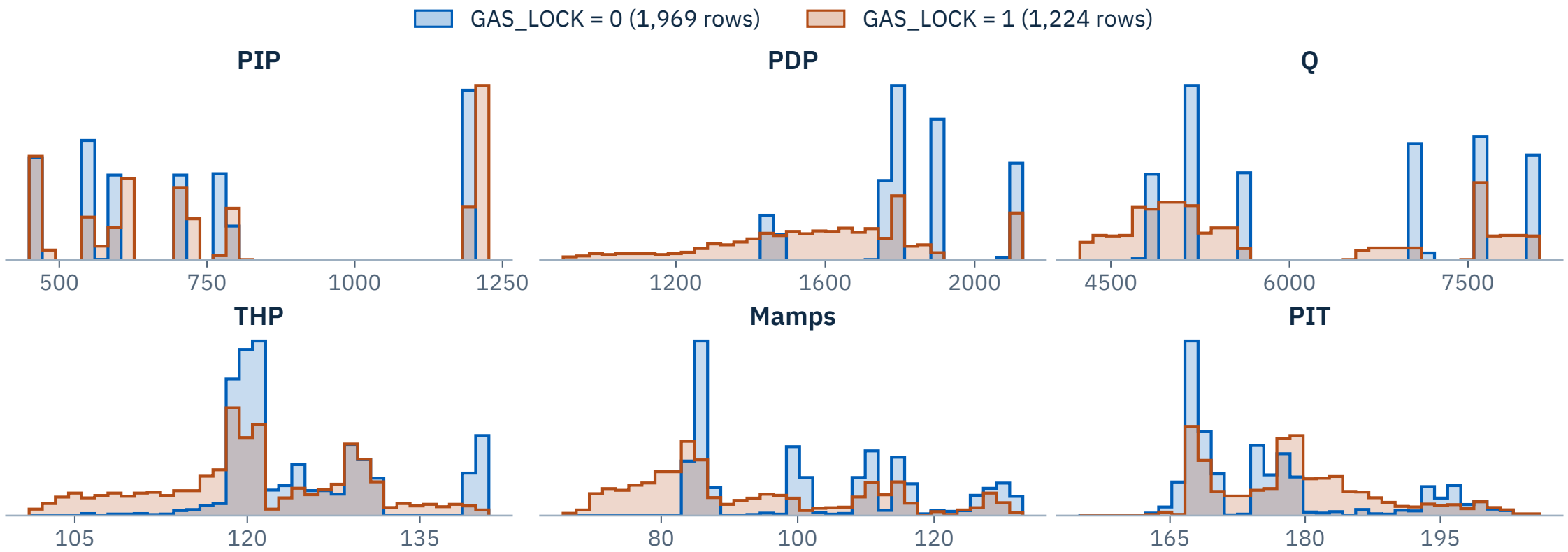
Supplied esp_train.csv. Counts are per row, not independent failure events. Missing sensor values require an explicit training-only treatment.

Gas-lock periods show up as ramps in the traces



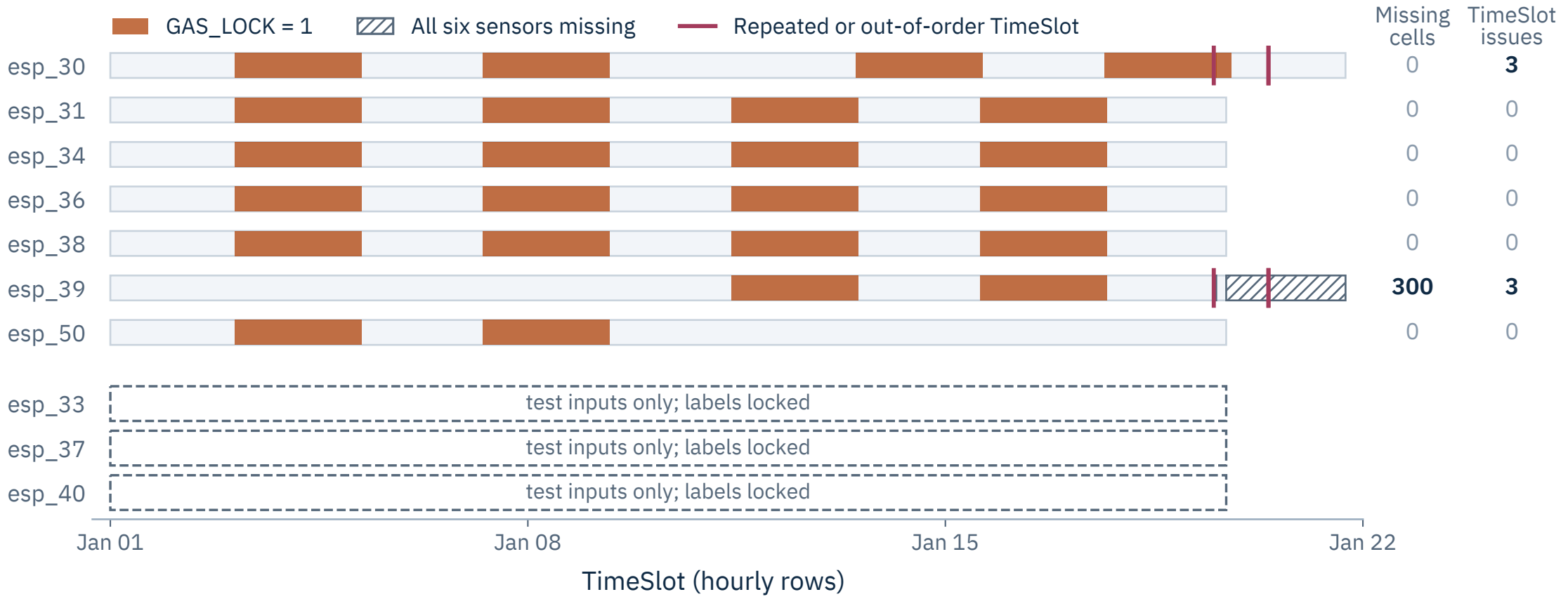
Computed from esp_train.csv: entities esp_30 and esp_39 in TimeSlot order; values as supplied, units undocumented.

Pooled histograms mix seven operating set-points



Computed from esp_train.csv: 3,193 rows with all six sensors present, pooled over seven entities.

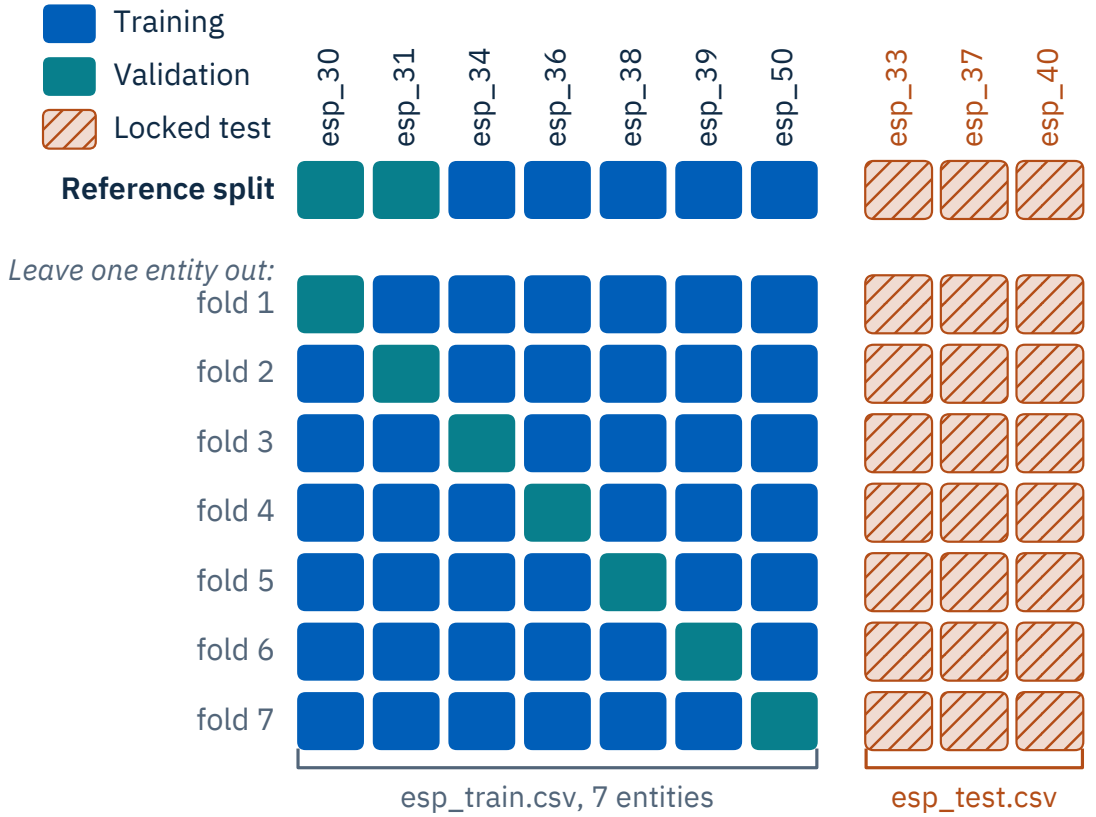
Missing rows and odd timestamps sit in two entities



Computed from esp_train.csv and the ENTITY and TimeSlot columns of esp_test.csv; the test label file was not read.

The ESP question is about unseen entities

- Can the model classify rows from an entity it has never seen?
- Group validation by ENTITY within the seven training entities
- Keep ENTITY and TimeSlot out of the numeric inputs
- Early warning is a separate task that needs a future target



Computed: GroupShuffleSplit(test_size=2, random_state=42) holds out esp_30 and esp_31; leave-one-entity-out repeats the question seven times.

Split by entity before fitting any preprocessing

```python
from sklearn.model_selection import GroupShuffleSplit

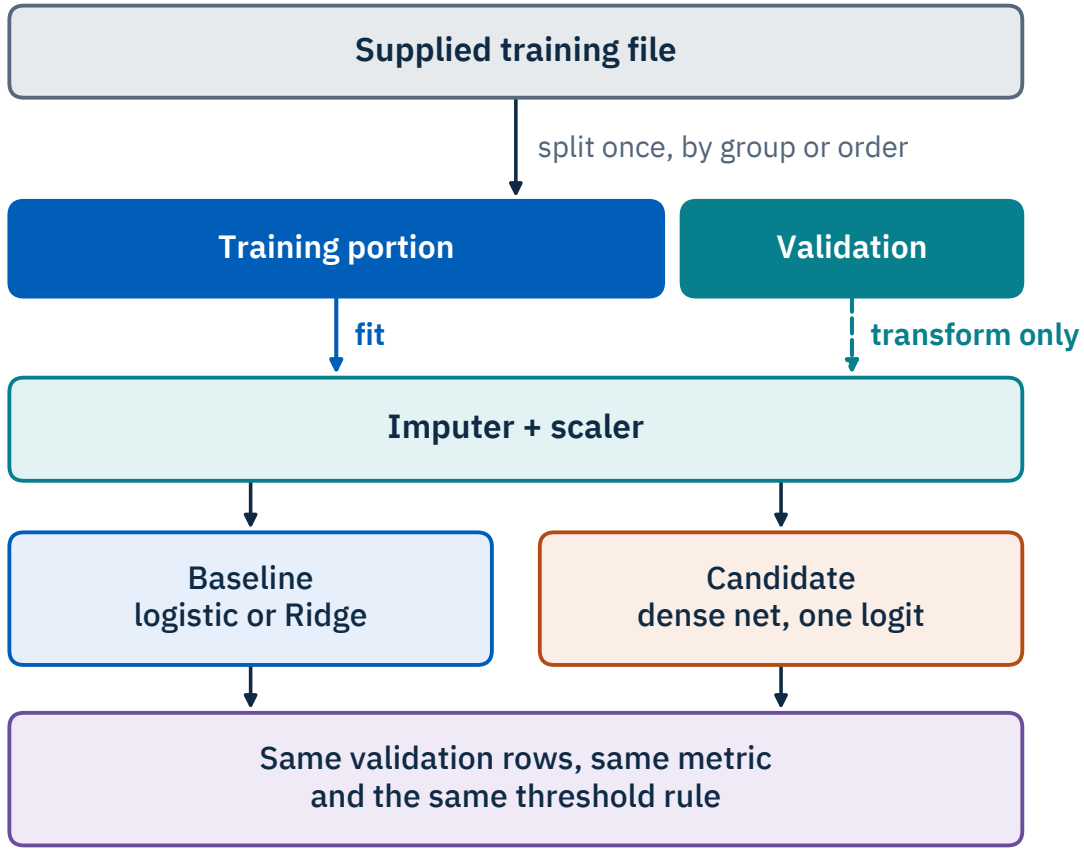
features = ["PIP", "PDP", "Q", "THP", "Mamps", "PIT"]
splitter = GroupShuffleSplit(
    n_splits=1, test_size=2, random_state=42
)
train_idx, val_idx = next(splitter.split(
    train_df[features], train_df["GAS_LOCK"],
    groups=train_df["ENTITY"]
))
val_ents = set(train_df["ENTITY"].iloc[val_idx])
assert not val_ents & set(train_df["ENTITY"].iloc[train_idx])
print(sorted(val_ents), len(train_idx), len(val_idx))
```

OUTPUT

```
['esp_30', 'esp_31'] 2295 948
```

Baseline and candidate share every other choice

- Baseline: median imputation, scaling, logistic regression
- Candidate: one hidden layer and a raw-logit loss
- Report gas-lock precision, recall and per-entity results
- Add average precision as a threshold-free check



Schematic. Split and fitted preprocessing are shared, so only the model differs.

The ESP final join must be checked

```
scored = test_df[["uuid"]].copy()
scored["probability"] = probabilities
scored = scored.merge(
    actual_df, on="uuid", how="left", validate="one_to_one"
)
assert len(scored) == len(test_df)
assert scored["GAS_LOCK"].notna().all()
```

Run this once, at milestone 4, after the configuration is locked.

⚠ A join that leaves missing labels means the files do not correspond: stop, do not drop rows.

PART 3 OF 7

03

Project 2: PE log-curve prediction

Regression with limited grouping metadata

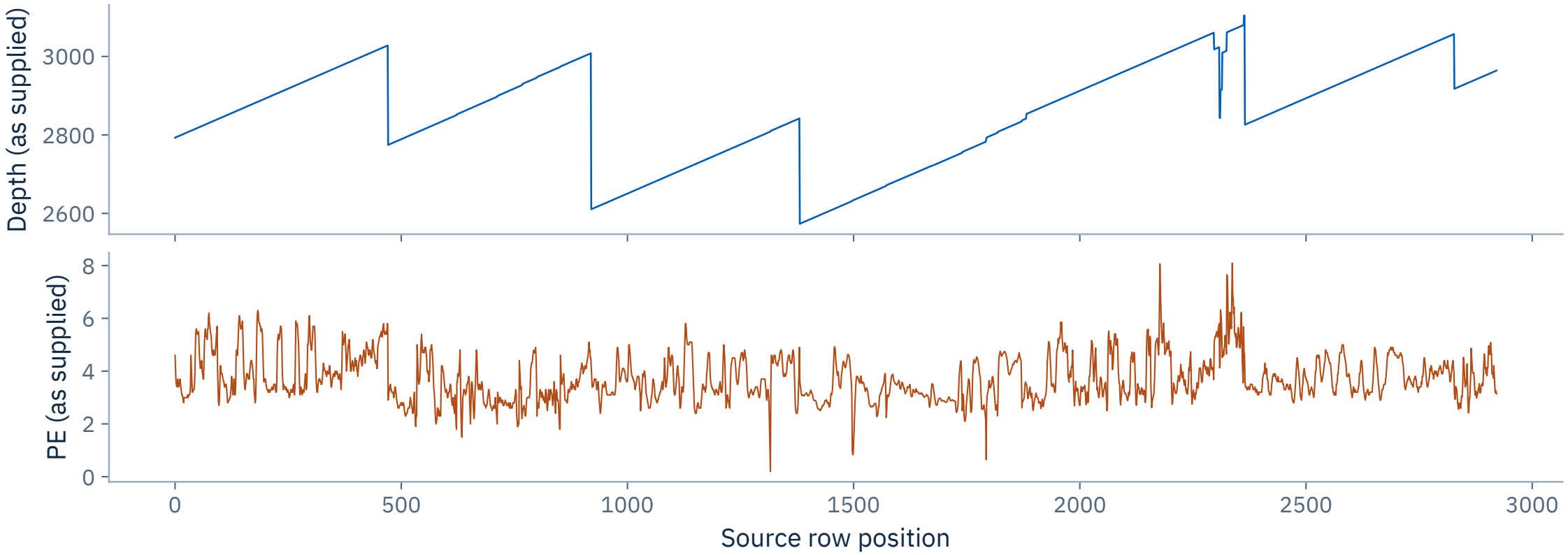
- Schema, depth resets and duplicated rows
- A well-log view of one depth block
- Correlations and the target distribution
- A contiguous-block validation fallback

Six inputs and PE, but no well identifier

Role	Columns	Known limitation
Predictors	Depth, GR, ILD_log10	Units and acquisition grouping need documentation
Predictors	DeltaPHI, PHIND, RELPOS	Use provided definitions when available
Training target	PE	Numeric regression target
Supplied evaluation	310 input rows and 310 PE labels	Label alignment has no explicit key

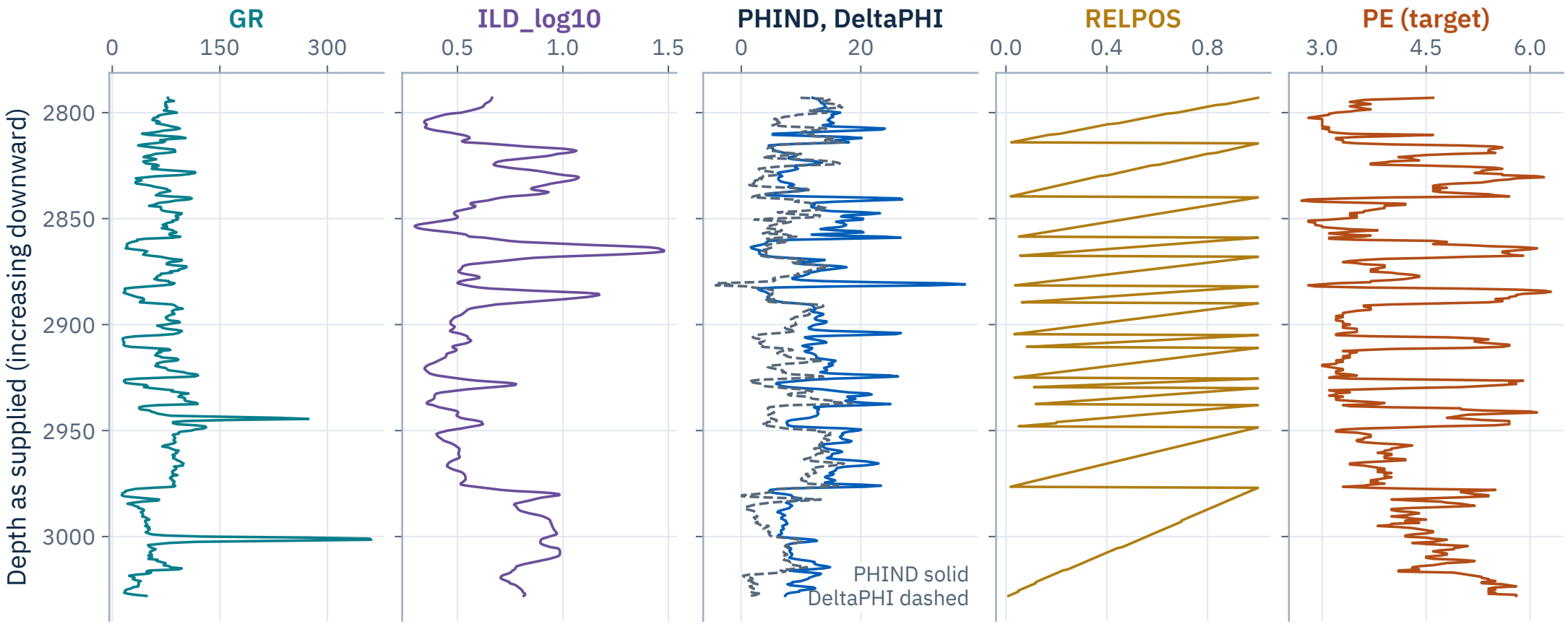
- Depth is supplied as a column, not proven to be measured depth in a single well.

Depth resets split the file into eight blocks



Supplied log training data, plotted in source-row order. The file lacks well identifiers, so changes in depth cannot alone establish well boundaries.

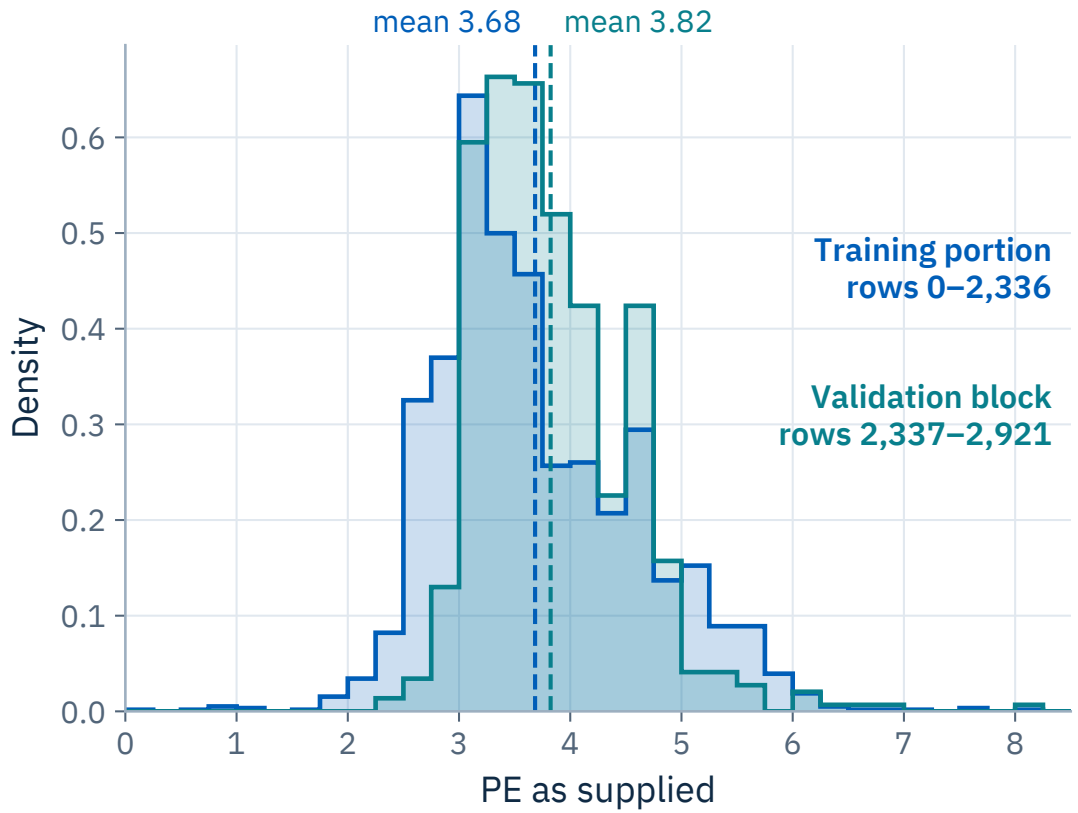
Read one depth block as a well log first



Computed from Log curve - Train.csv: source rows 0–470, the first block of increasing Depth; curves as supplied, units undocumented; well identity is not given.

PE follows PHIND and ILD; validation PE runs higher

	Depth	GR	ILD	DeltaPHI	PHIND	RELPOS	PE
Depth		−.07	.15	−.04	−.13	−.02	**.31**
GR	−.07		−.16	.20	.27	−.14	**−.27**
ILD	.15	−.16		−.11	−.56	.09	**.45**
DeltaPHI	−.04	.20	−.11		−.23	.05	**.01**
PHIND	−.13	.27	−.56	−.23		−.05	**−.59**
RELPOS	−.02	−.14	.09	.05	−.05		**.01**
PE	**.31**	**−.27**	**.45**	**.01**	**−.59**	**.01**	



Computed from Log curve - Train.csv. Left: Pearson correlations over all 2,922 rows (ILD = ILD_log10; descriptive, rows are not independent). Right: PE in the training portion and the validation block of the reference split.

A contiguous source-row block is the honest fallback

- Hold out a contiguous block in source-row order
- Fit input and target scalers on the remaining rows only
- Compare with the training mean and a Ridge baseline
- Asset-level generalization stays unverified without well IDs



Computed from Log curve - Train.csv: the reference split keeps rows 0–2,336 for training and holds out rows 2,337–2,921.

Regression evaluation stays in target units

Three habits turn one PE score into evidence about where the model works.

Invert any scaling

- Report MAE and RMSE in PE units
- Target scaler fitted on training rows

Plot against the truth

- Observed and predicted with an identity line
- Residuals against depth and key inputs

Find where errors live

- Depth blocks, ranges, sparse regions
- R^2 can be negative

◆ Predicting the training mean gives $R^2 = -0.04$ on the validation block, whose mean PE is 3.82, not 3.68.

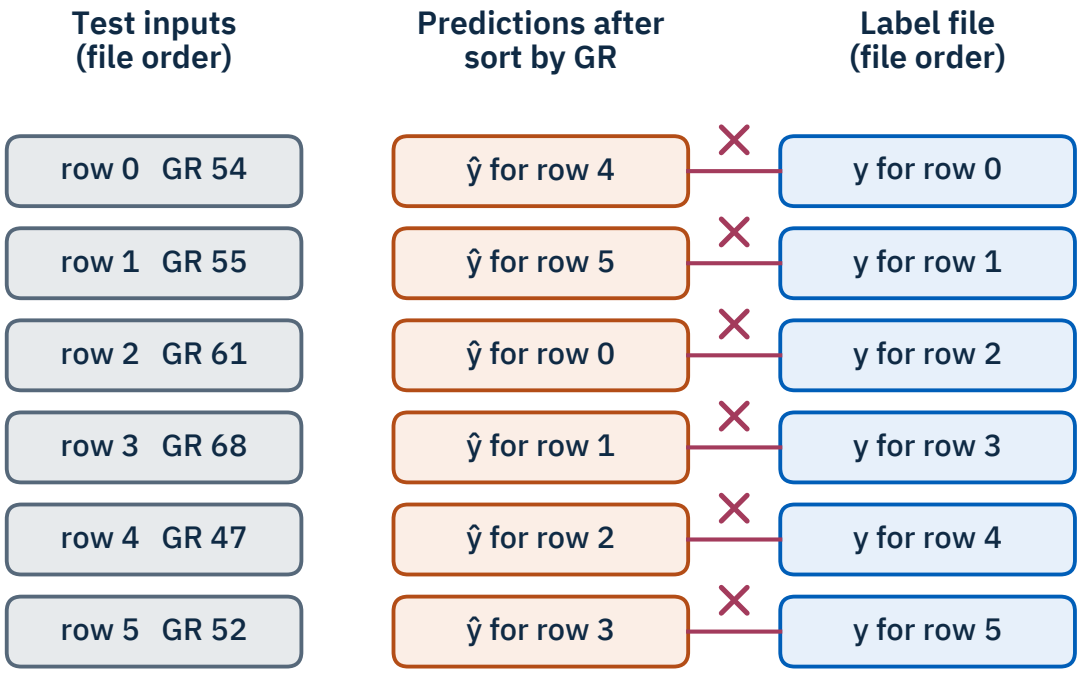
PRACTICE

4 min · Pairs

Positional labels require row-order discipline

You sort the PE test inputs by GR to inspect outliers, predict, and score against the unchanged label file.

1. What happens to the score?
2. Why does the row-count check still pass?
3. How should the evaluation preserve alignment?



Row counts match (310 = 310), so the length check passes,

yet every prediction is scored against another row's label.

Schematic with illustrative GR values.

PART 4 OF 7

04

Project 3: sensor-signal classification

Anonymous numeric inputs with explicit record keys

- Schema and the label convention
- Profiles, class ranges and near-copy columns
- Duplicates, uuid order and split design
- Modelling and evaluation with stated limits

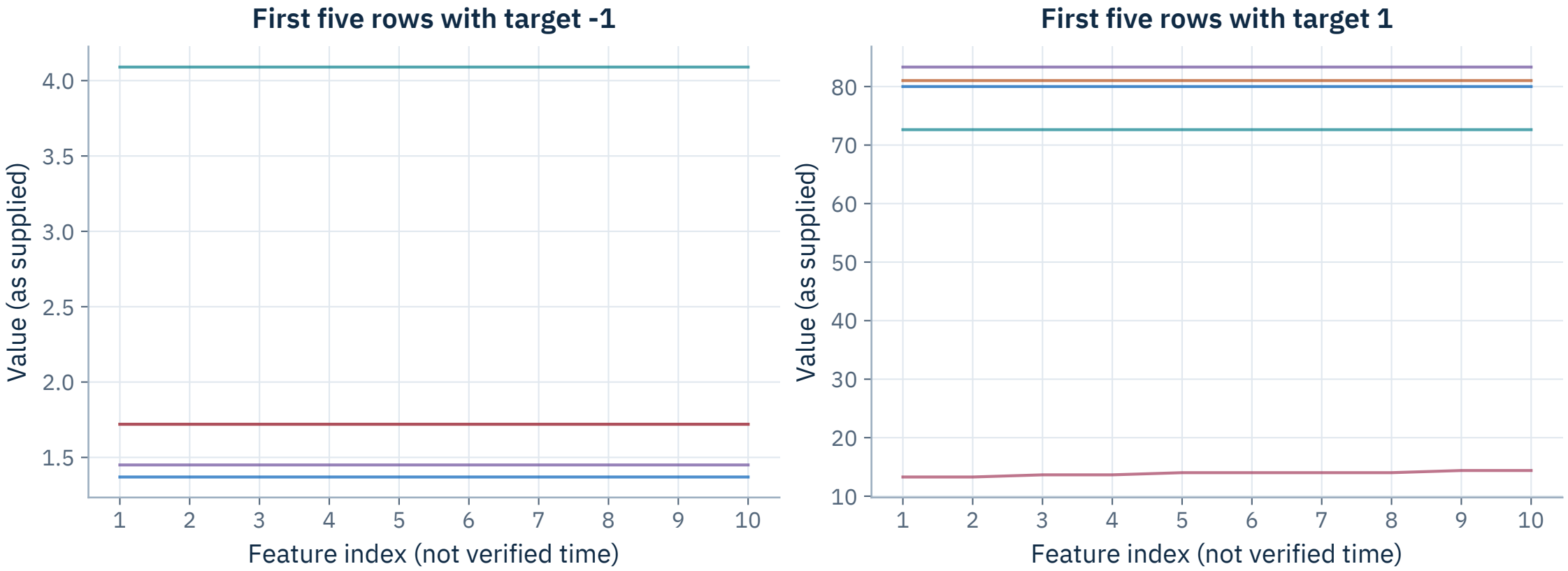
Ten anonymous columns, a key and ±1 labels

```
target = train_df["target"]
assert set(target) == {-1, 1}
y = (target == 1).astype(int)
X = train_df[[f"col{i}" for i in range(1, 11)]]
assert train_df["uuid"].is_unique
```

- Ten predictors: col1 through col10
- uuid identifies records and joins test labels
- Labels are −1 and 1, not 0 and 1
- Map them explicitly before a binary-logit loss

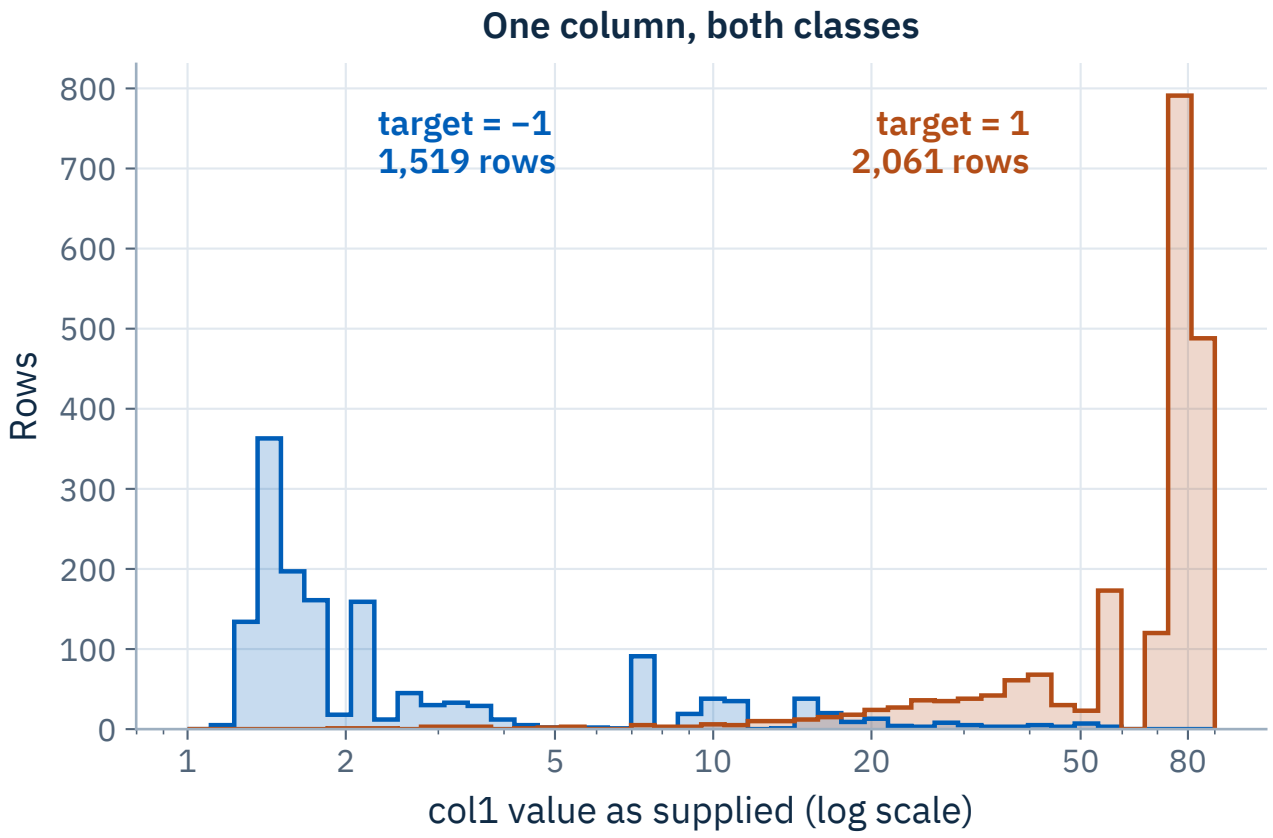
⚠ BCEWithLogitsLoss accepts a −1 target without error and silently optimizes the wrong objective.

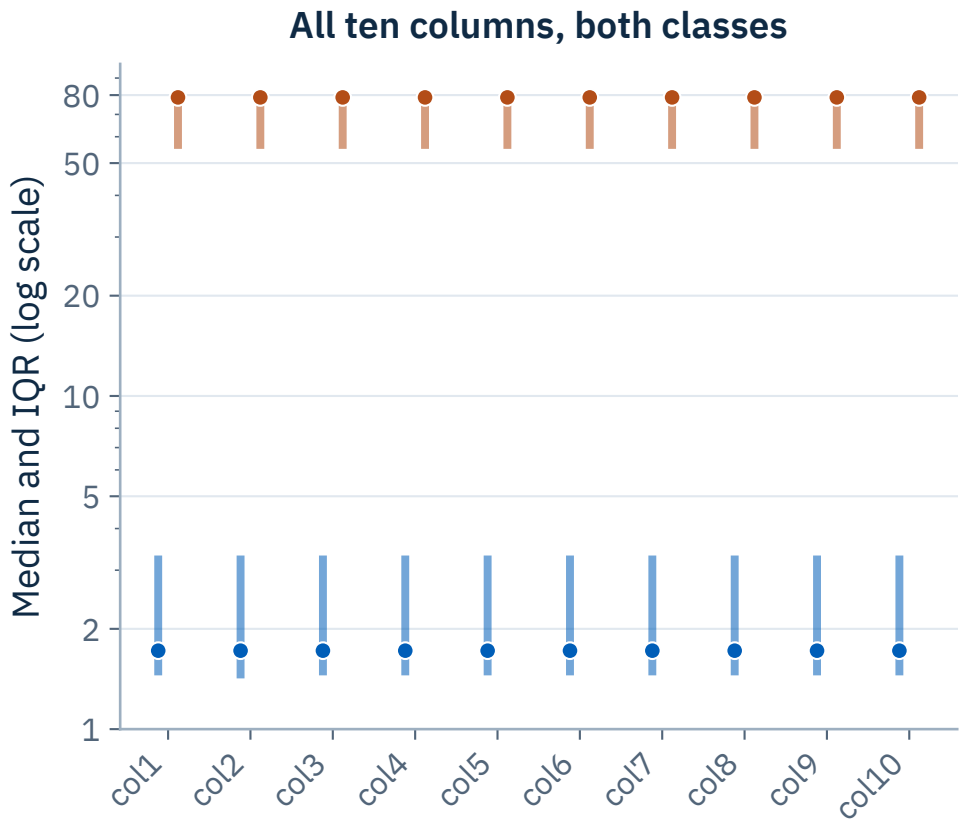
Inspect the profiles before choosing an architecture



Selected rows from the supplied sensor training file. The horizontal axis is feature index, not verified elapsed time.

The two classes occupy different value ranges

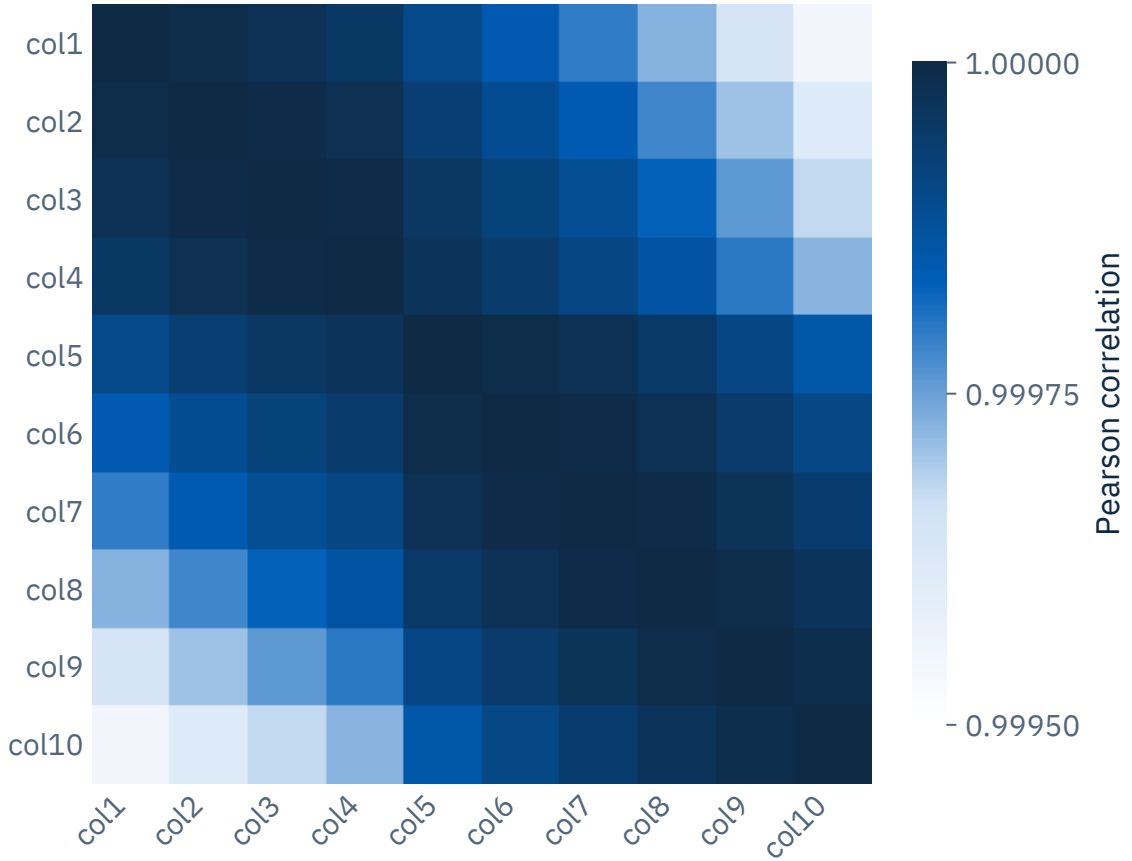




Computed from Sensor_train.csv (3,580 rows). Column index is a name, not a verified time axis.

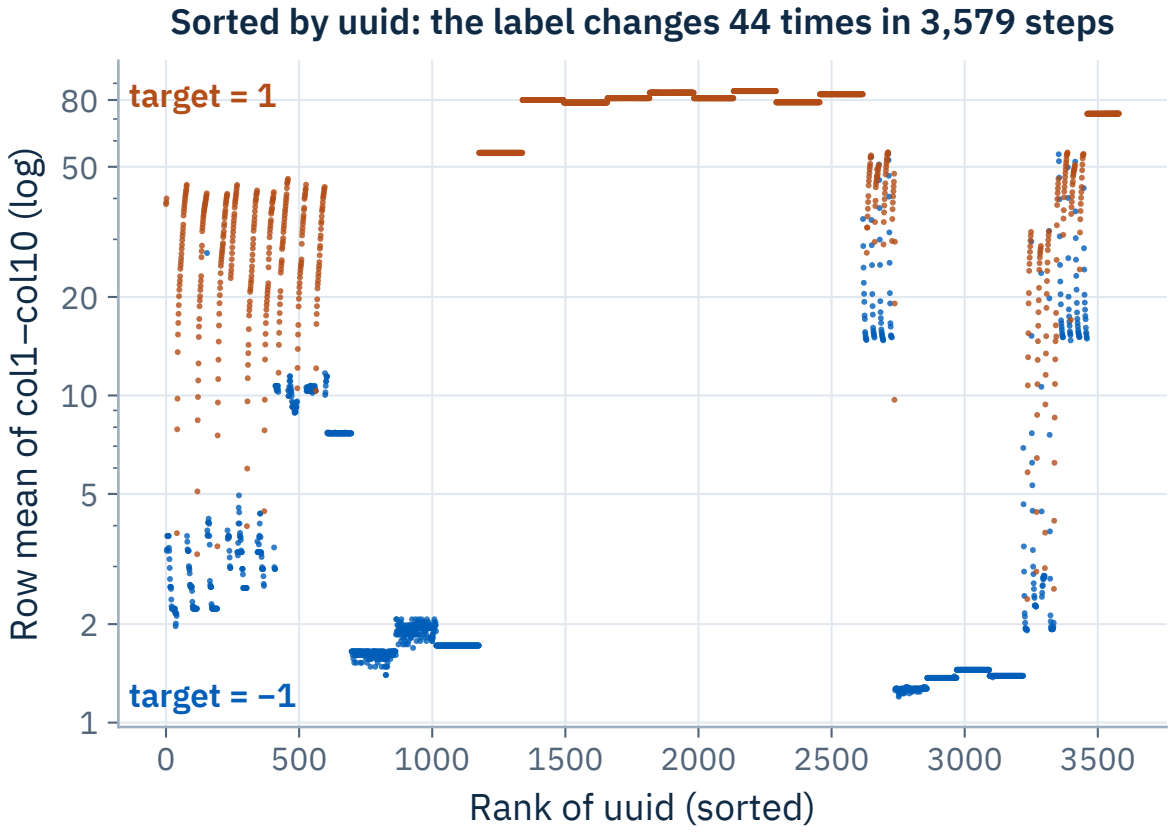
The ten columns are near-copies of one another

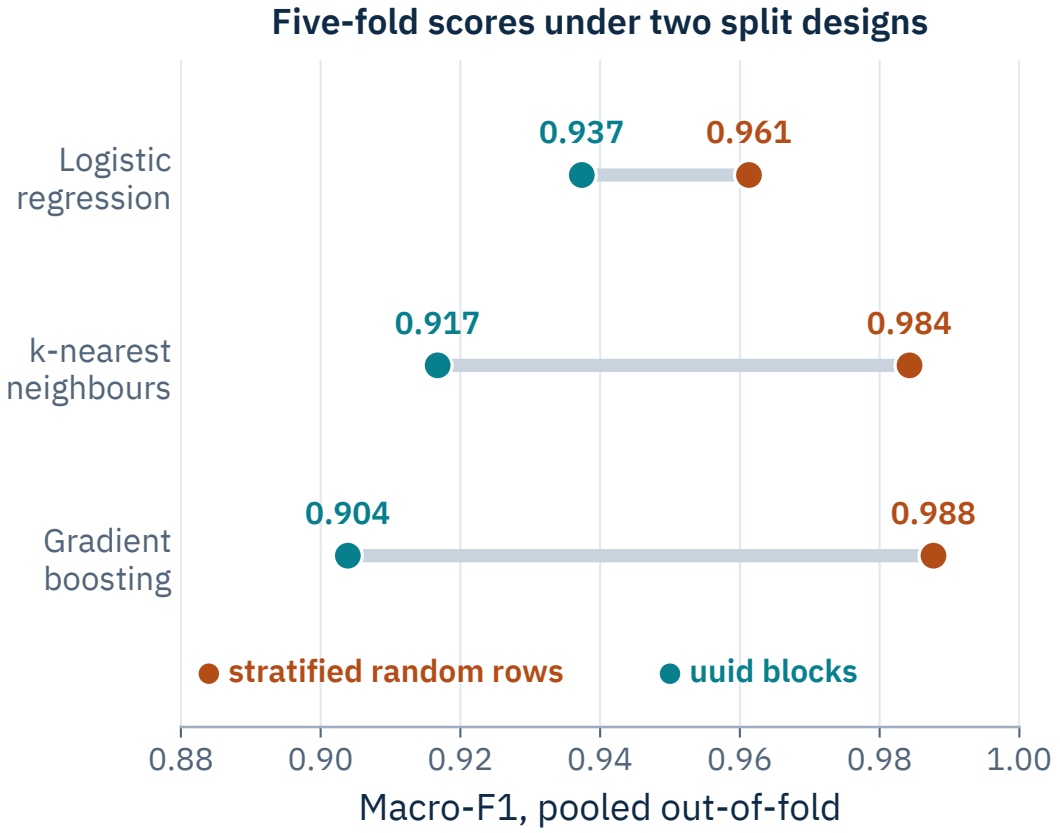
- Every pairwise correlation is at least 0.9995
- 65% of rows hold ten identical values, in both classes
- Treat the columns as one signal plus small deviations
- No data dictionary supports reading the index as time



Computed from Sensor_train.csv: Pearson correlations between col1–col10 over 3,580 rows.

Neighbouring uuids hold near-identical records





Computed from Sensor_train.csv: row means in uuid order; macro-F1 from stratified 5-fold versus five contiguous uuid blocks, scaling fitted inside folds, threshold 0.5.

Model the sensor data with its caveats stated

Baseline

- Train-only scaling, logistic regression
- Threshold chosen on validation

Candidate

- Small dense network, one raw logit
- Same split and preprocessing

Evaluation

- Macro-F1 and per-class metrics
- Stratified split as a documented fallback

Robustness

- Repeat with contiguous uuid blocks
- Report both splits side by side

◆ The supplied test is spread through the training uuid range, so it cannot show performance on a new range.

LABORATORY

35 min

Sprint 1: write the data contract and audit

- ☐ Form a team of two or three; open your project's student notebook
- ☐ Notebook sections 1–2: inspect the file and write the prediction contract
- ☐ Audit missing values, duplicates, keys, groups and order with the checklist
- ☐ Sections 3–4: check the split and the training-only preprocessing

NOTEBOOKS

`exercises/01_esp_gas_lock/ESP_Gas_Lock_Student.ipynb`

`exercises/02_log_curves/Log_Curve_Prediction_Student.ipynb`

`exercises/03_sensor_signals/Sensor_Signals_Student.ipynb`

EVIDENCE TO SUBMIT

- One-page data contract
- Audit table with a decision per finding
- Split policy and what it cannot show

MILESTONE 1 OF 4 · 11:20

Checkpoint 1: the data contract

Show

- Target, row unit, excluded columns
- Audit findings with a decision each
- The split policy in one sentence

Instructor checks

- No test labels opened
- The split respects groups or order
- Limits written down, not implied

✓ Proceed when

- The split is fixed and saved
- Preprocessing fits training rows only

◆ About two minutes per team: start with the split sentence, it reveals most problems.

PART 5 OF 7

05

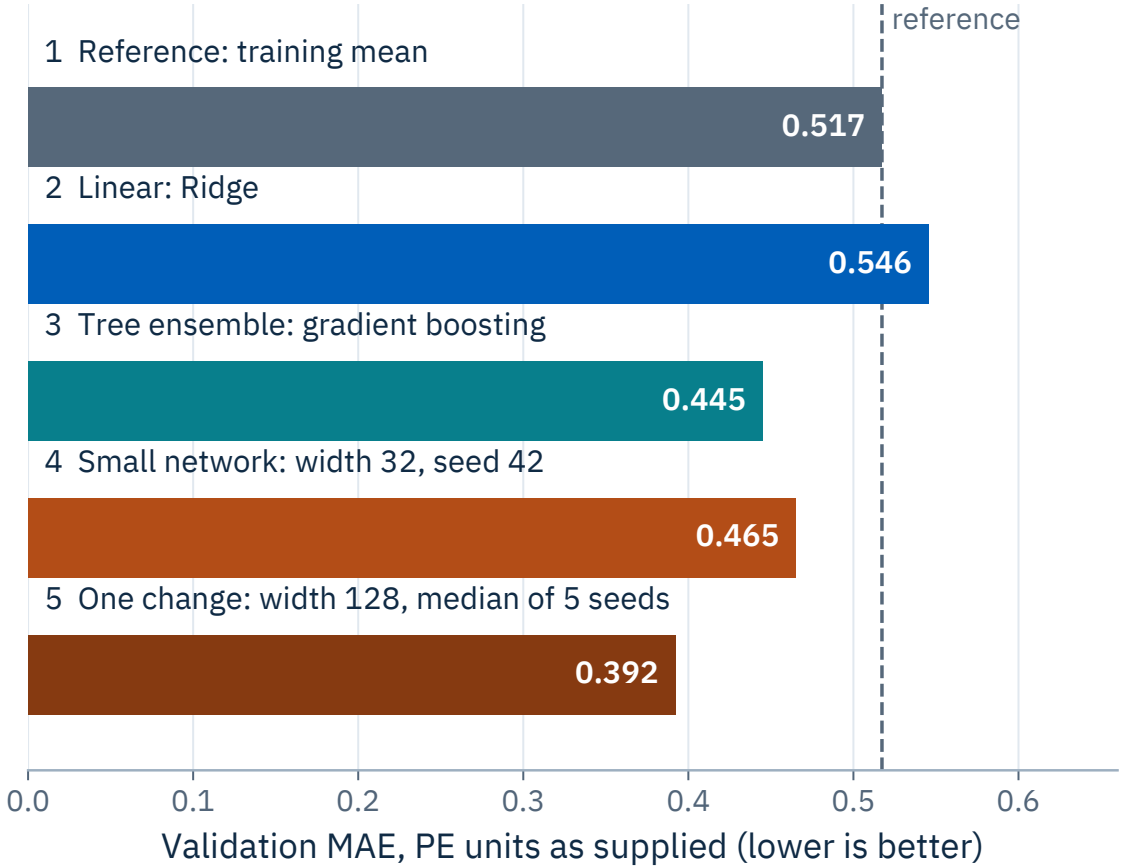
Baselines and experiments

A tabular-NN playbook for a few thousand rows

- Baselines first, including a tree ensemble
- Network size, scaling, missing values, early stopping
- After lunch: one controlled change over several seeds
- Candidate changes: class weights, loss, width

The playbook starts with the cheapest honest model

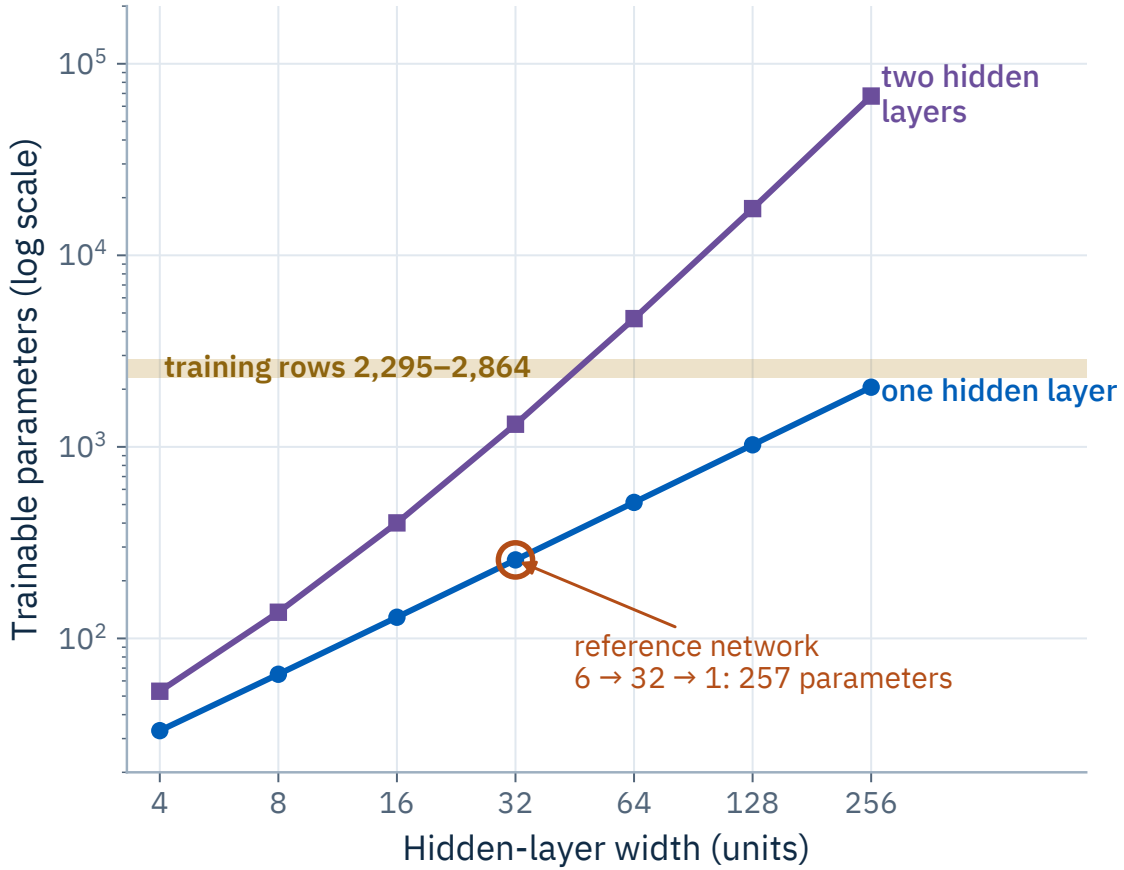
1. **Reference**: training mean or class prevalence
2. **Linear**: logistic or Ridge regression
3. **Tree ensemble**: gradient boosting
4. **Small network**: one hidden layer
5. **One change**: a controlled experiment



Computed on the PE reference split (source rows 0–2,336 train, 2,337–2,921 validation).

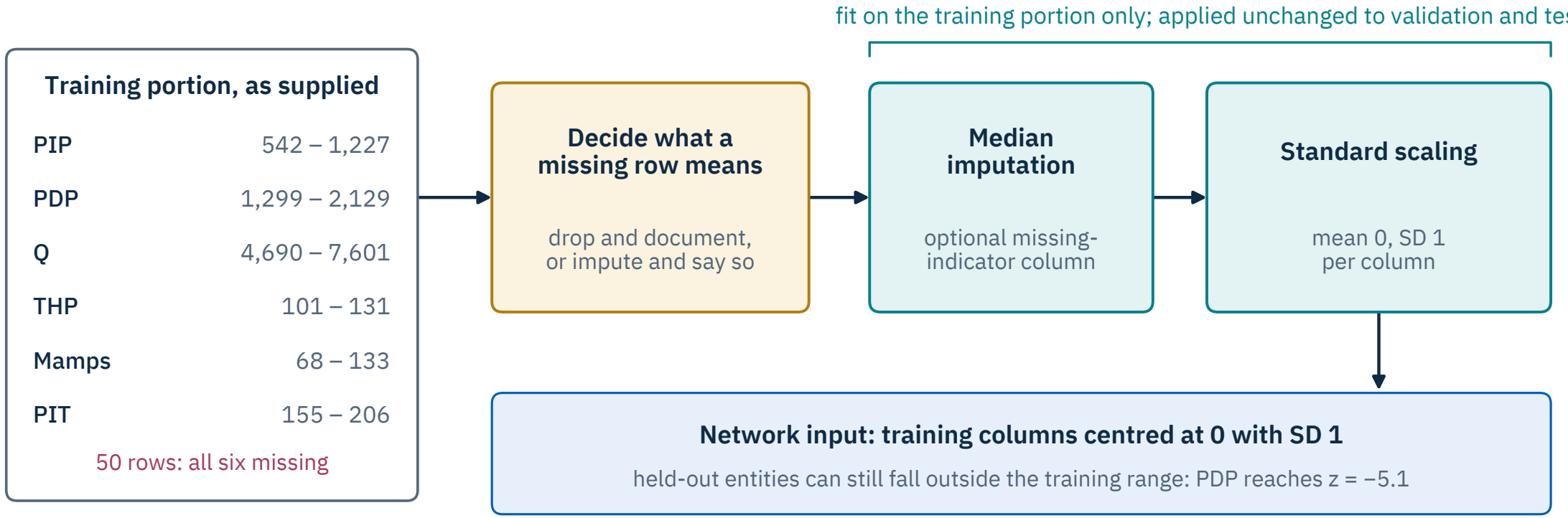
A few thousand rows call for a small network

- Start with one hidden layer of 16–64 units
- 6 → 32 → 1 has 257 parameters; two layers of 64 have 4,673
- Parameter count is not a hard limit; validation decides
- Width is cheap to test; depth adds tuning burden



Exact parameter counts for dense networks with 6 inputs and one output; the band spans the training rows of the three reference splits.

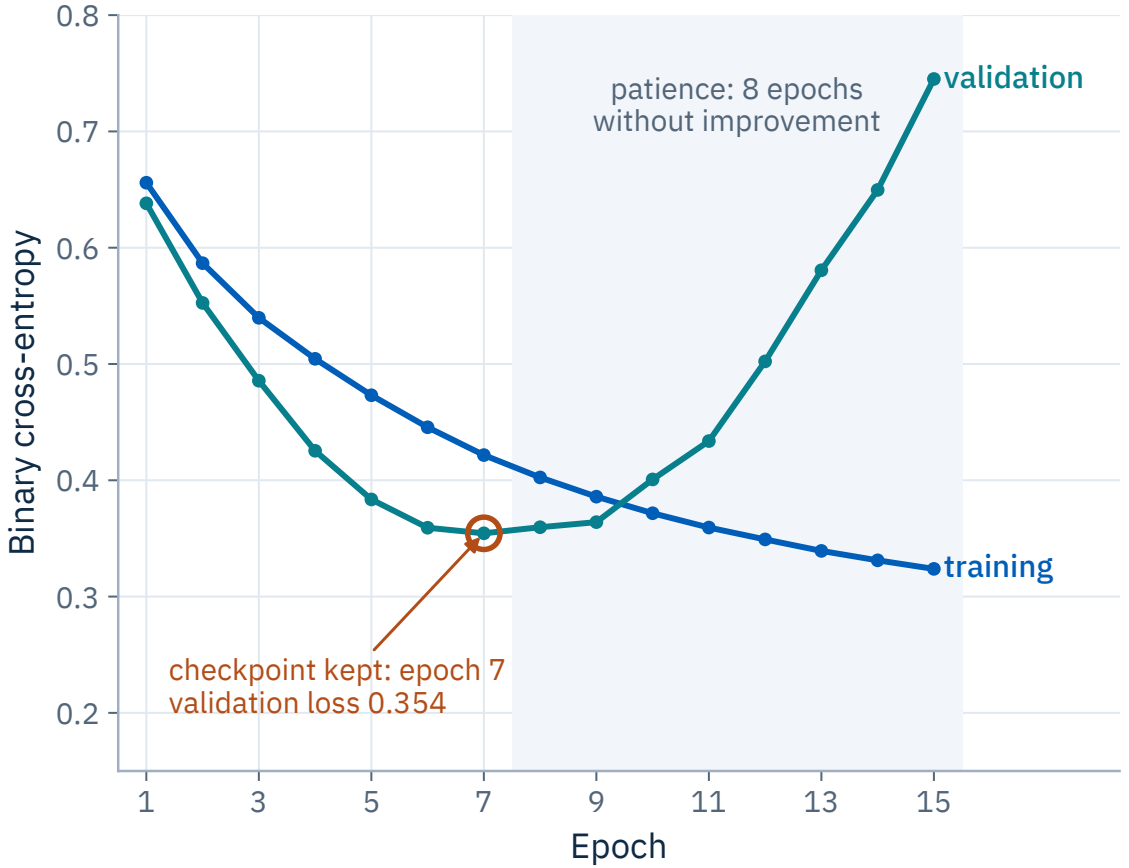
Scale inputs and decide what a missing row means



Schematic; value ranges computed from the training portion of the reference ESP split in esp_train.csv.

Early stopping keeps the best validation checkpoint

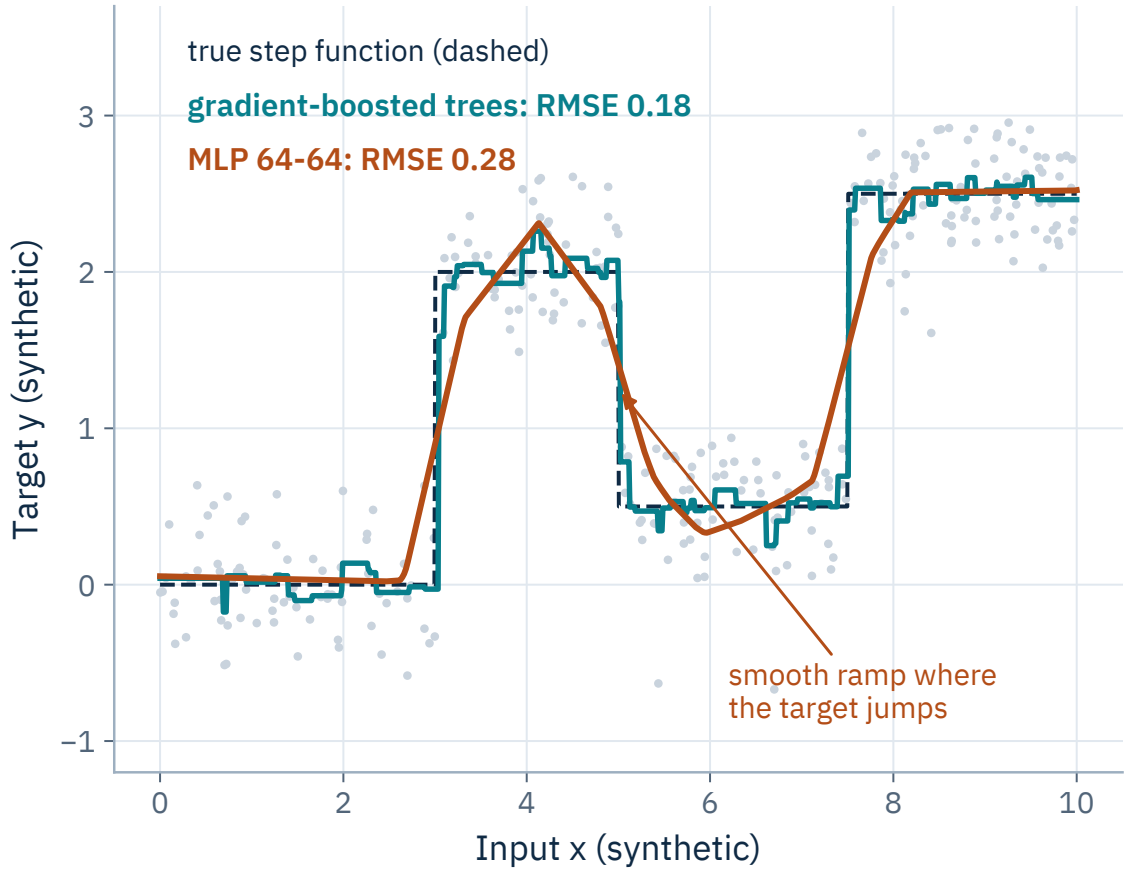
- Validation loss is lowest at epoch 7, then climbs
- The loop copies that state; patience is 8 epochs
- Restore the copy before evaluating or saving
- Training loss keeps falling: the gap is the warning



Computed: curriculum/capstone.py on the ESP project (width 32, AdamW, lr 0.001, seed 42); validation = esp_30 and esp_31.

Tree ensembles are often strong on tabular data

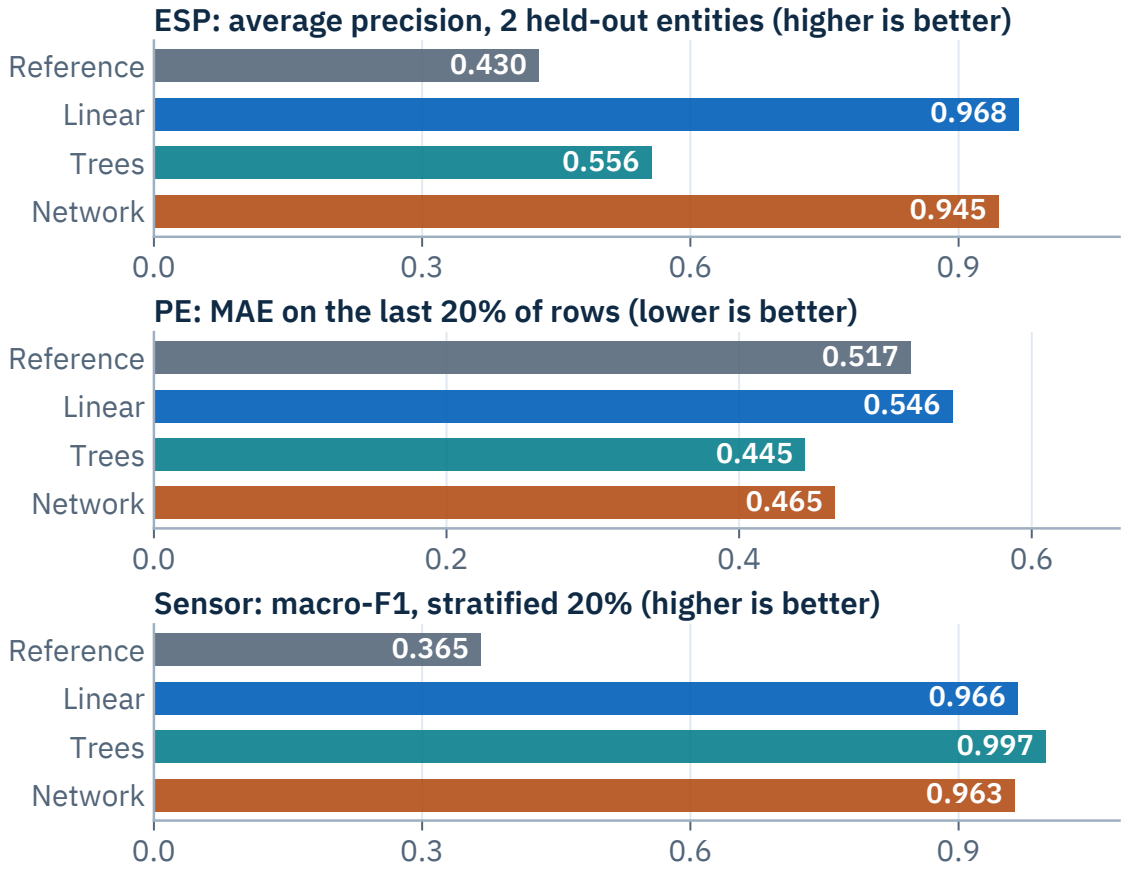
- A 45-dataset benchmark found trees ahead on medium-sized tables
- Networks favour smooth functions; trees fit jumps readily
- Uninformative columns hurt MLPs more than trees
- Columns carry meaning; trees split along them



Synthetic example, seed 0: 300 noisy samples of a step function; RMSE against the true function.

Treat a tree ensemble as a second baseline

- Fit gradient-boosted trees on the same rows; they accept missing values
- Trees win on PE and on sensor rows, but fail on held-out ESP pumps
- Rank models per project and per split, not by reputation
- The network does not need to win; a clear negative result counts



Computed on the reference splits (training files only). Network: width 32, seed 42. Sensor thresholds chosen on validation.

LABORATORY

40 min

Sprint 2: fit the baselines and a first network

- [ ] Section 5: run the reference baseline and network on your split
- [ ] Add a no-skill reference and a tree ensemble on the same rows
- [ ] Section 6: read the learning curve and note the selected epoch
- [ ] Save the run folder; record the command, seed and versions

NOTEBOOKS

`exercises/01_esp_gas_lock/ESP_Gas_Lock_Student.ipynb`

`exercises/02_log_curves/Log_Curve_Prediction_Student.ipynb`

`exercises/03_sensor_signals/Sensor_Signals_Student.ipynb`

EVIDENCE TO SUBMIT

- Score table: reference, linear, tree, network
- Learning curve with the selected epoch
- The exact command that reproduced it

MILESTONE 2 OF 4 · 12:30

Checkpoint 2: a reproducible baseline

Show

- Score table, reference first
- Learning curve and the selected epoch
- The command that reproduced it

Instructor checks

- Same validation rows for every model
- Medians fitted on training rows only
- Threshold chosen on validation

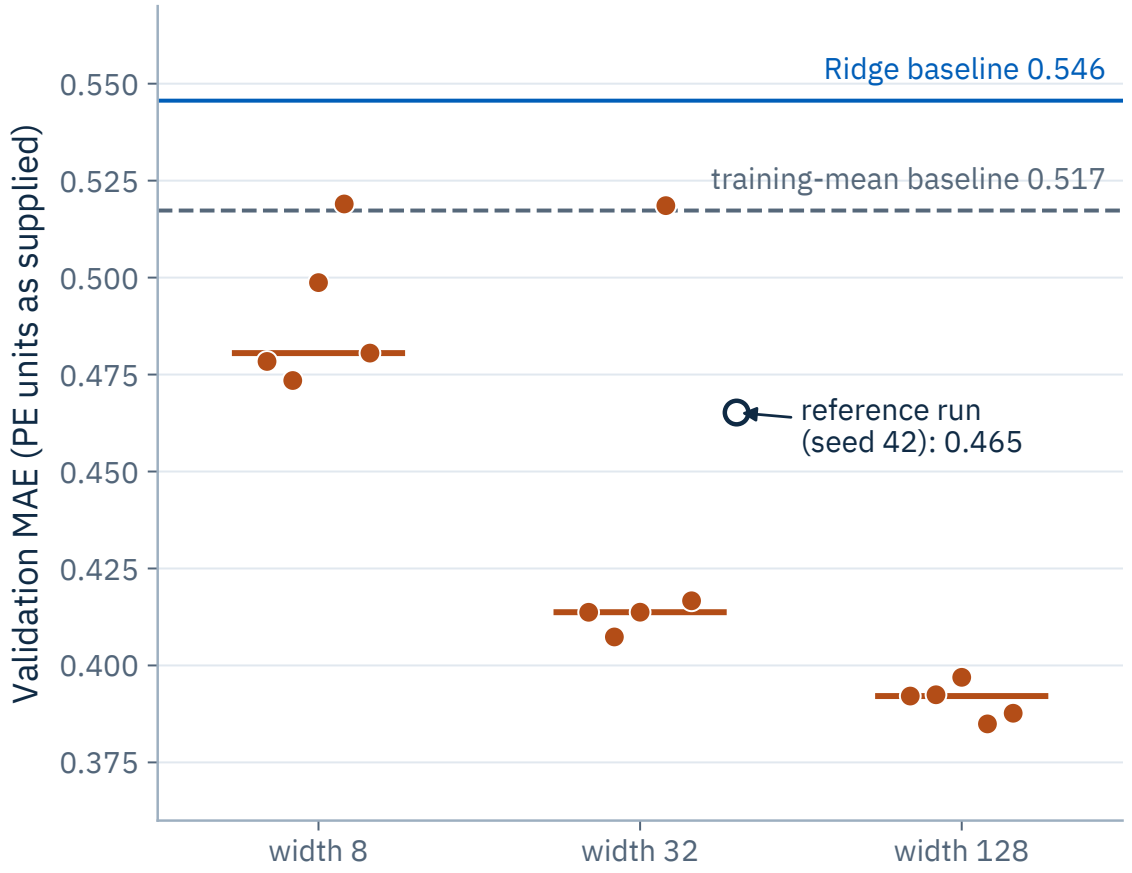
✓ Proceed when

- A fresh process gives the same numbers
- The team can say which model leads, by how much

◆ A solid baseline without a network passes this gate; a network without a baseline does not.

A controlled experiment changes exactly one thing

- Write the hypothesis and expected direction first
- Hold split, preprocessing, budget and seeds fixed
- Repeat each setting over at least three seeds
- Here, one seed at width 32 scored like width 8

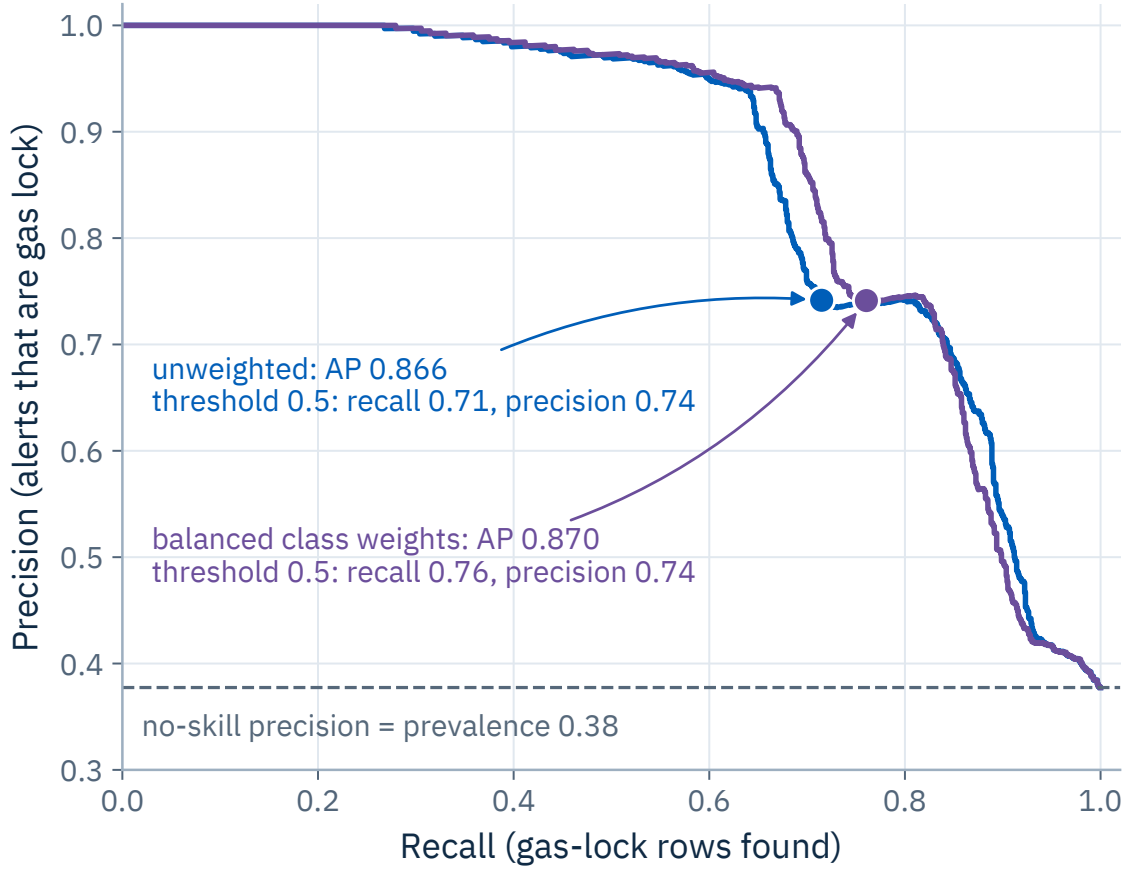


Computed: reference training loop on the PE project, seeds 0–4 per width (bars mark medians); baselines fitted on the same training rows.

Class weights move the operating point, not the ranking

$$L = -\frac{1}{n}\sum_i [w_1\, y_i \log p_i + w_0\,(1-y_i)\log(1-p_i)]$$

- Balanced weights set $w_c = n/(2n_c)$ for each class c
- ESP: AP 0.866 → 0.870; recall at 0.5 rises 0.71 → 0.76
- A threshold chosen on validation does a similar job



Computed from esp_train.csv: logistic regression, leave-one-entity-out predictions pooled over seven entities.

Each project needs a matching loss and metric

Project	Output	Loss	Evaluation
ESP	One logit	Binary cross-entropy with logits	PR metrics, confusion counts
PE	One numeric value	MSE or Huber	MAE, RMSE, residuals
Sensor	One logit	Binary cross-entropy with logits	Macro-F1, per-class metrics

Train on the loss, select on validation loss or metric, and report the metric in target units.

- Probabilities and thresholds belong to the decision step, not to the training loss.

Good first experiments for each project

ESP

- Class weights on versus off
- All seven entity folds, not one split

PE

- Width 32 versus 128, five seeds
- MSE versus Huber loss

Sensor

- Stratified versus uuid-block split
- One row-mean feature versus ten columns

◆ One change, a written hypothesis, and a decision rule set before the runs.

LABORATORY

40 min

Sprint 3: run one controlled experiment

- [ ] Write the hypothesis, the change and the decision rule first
- [ ] Run both settings over at least three seeds; change nothing else
- [ ] Compare the difference with the seed spread
- [ ] Decide, then write the locked configuration down

NOTEBOOKS

`exercises/01_esp_gas_lock/ESP_Gas_Lock_Student.ipynb`

`exercises/02_log_curves/Log_Curve_Prediction_Student.ipynb`

`exercises/03_sensor_signals/Sensor_Signals_Student.ipynb`

EVIDENCE TO SUBMIT

- Hypothesis, result and decision in three sentences
- Seed-by-seed table for both settings
- Locked configuration: width, lr, seed, epochs, threshold rule

MILESTONE 3 OF 4 · 14:20

Checkpoint 3: hypothesis, result, decision

Show

- The hypothesis written before the runs
- Both settings over the same seeds
- The locked configuration in writing

Instructor checks

- Exactly one change between settings
- The difference is compared with the spread
- The test file is still closed

✓ Proceed when

- Configuration and command are frozen
- A null result is recorded as a result

◆ After this gate only analysis and presentation may change.

PART 6 OF 7

06

Error analysis

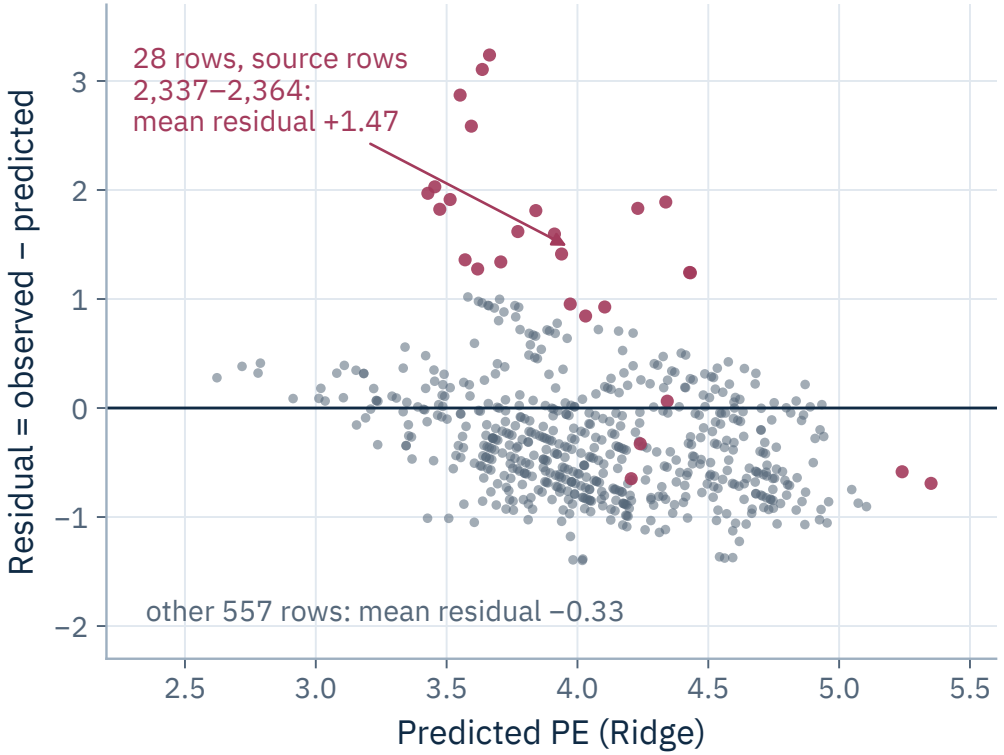
What the score hides, and how to find it

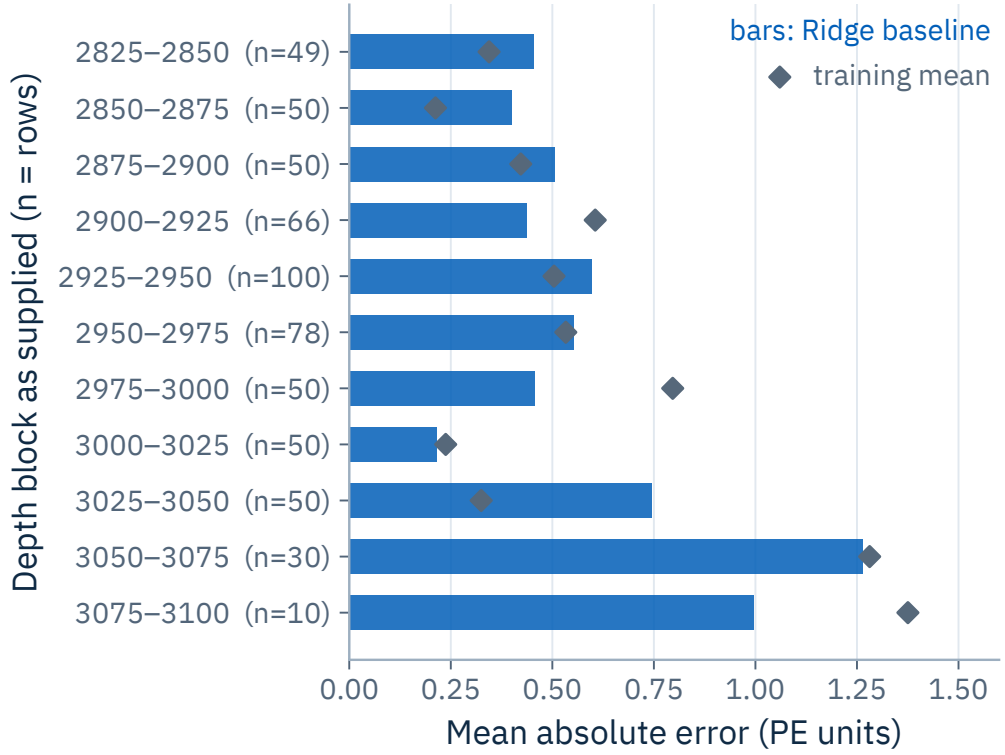
- Residuals against predictions and by depth block
- Errors by held-out entity
- Confusion by class
- Calibration of predicted probabilities

ERROR ANALYSIS

Residuals reveal a biased 28-row block in PE

Ridge baseline fitted on source rows 0–2,336 and scored on the validation block.

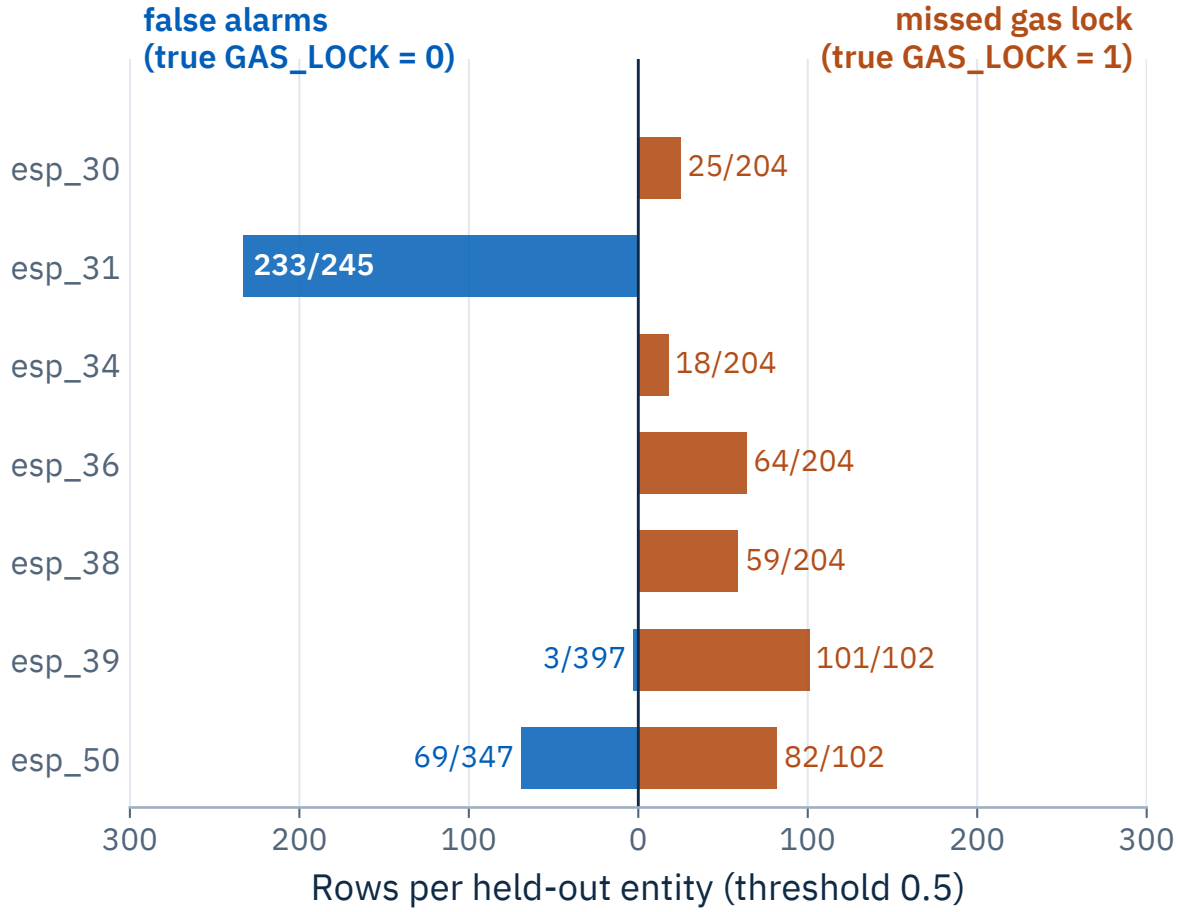




Computed from Log curve - Train.csv. Left: residuals against predictions. Right: mean absolute error by 25-unit Depth block (bars: Ridge; diamonds: training mean).

Two held-out entities fail in two different ways

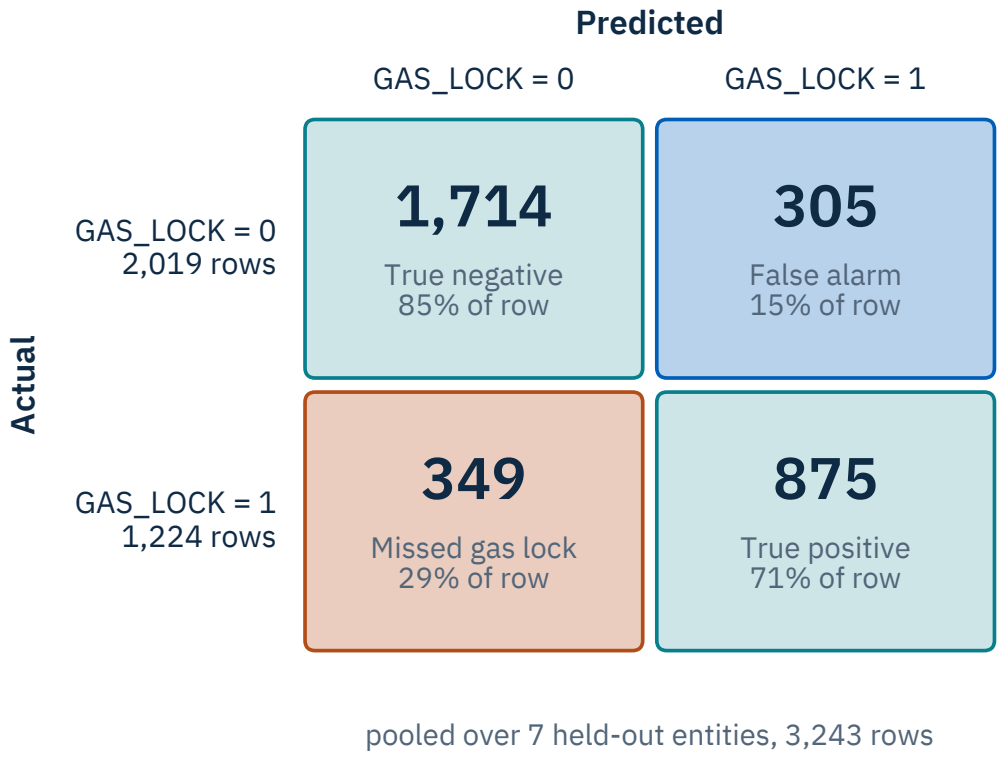
- esp_31: every gas-lock row found, 233 of 245 normal rows flagged
- esp_39 and esp_50: most gas-lock rows missed
- Pooled, these failures average into one number
- Each failure suggests a different next test



Computed from esp_train.csv: logistic baseline, leave-one-entity-out, threshold 0.5.

A confusion matrix separates misses from false alarms

- Rows are actual classes; columns are predictions
- 349 of 1,224 gas-lock rows missed (29%)
- 305 of 2,019 normal rows flagged (15%)
- Decide which error costs more before moving the threshold

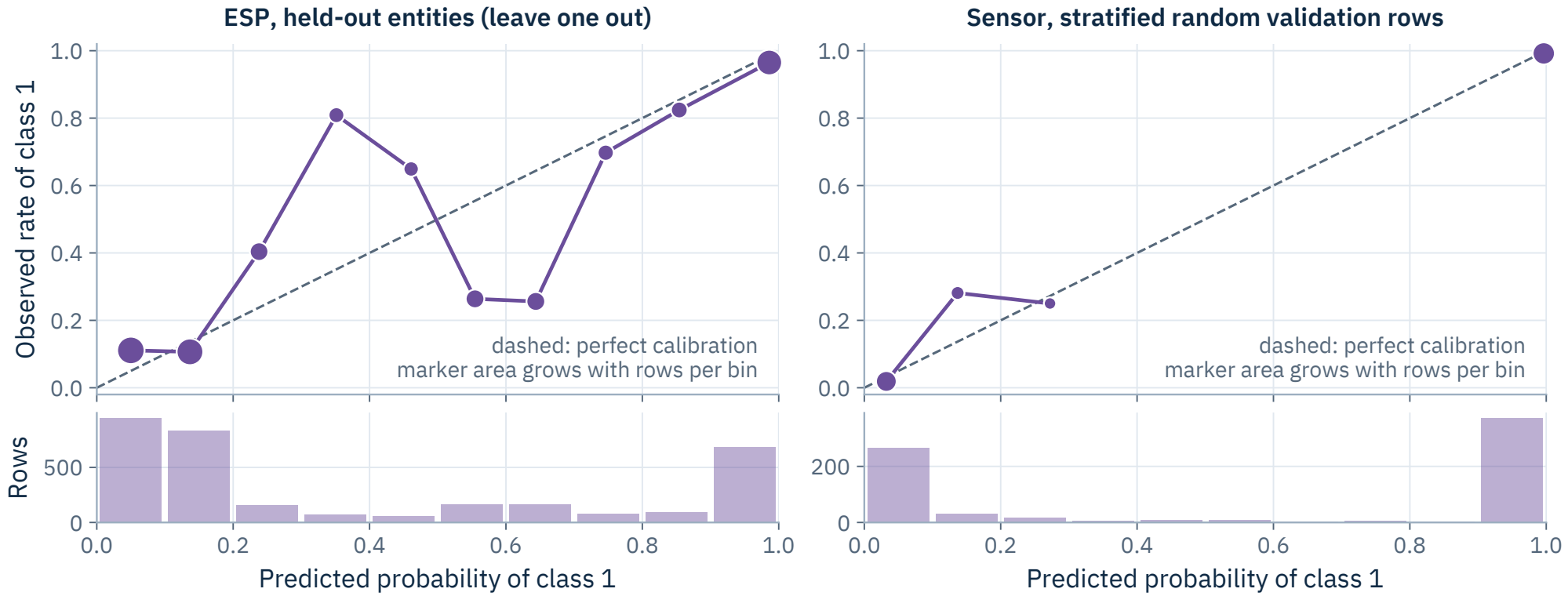


Computed from esp_train.csv: logistic baseline, leave-one-entity-out predictions pooled over seven entities, threshold 0.5.

ERROR ANALYSIS

Calibration shows whether probabilities can be trusted

Group predictions into bins and compare the average prediction with the observed rate.



Computed: logistic baselines. Left: esp_train.csv, leave-one-entity-out (3,243 rows). Right: Sensor_train.csv, reference stratified validation rows (716). Bins with fewer than 10 rows omitted.

PRACTICE

10 min · Teams

An error review asks what the score hides

1. Choose one failed example or slice and inspect its inputs, target and prediction
2. Decide: label noise, missing information, model limitation, or shift?
3. Propose one experiment that could tell the explanations apart

◆ Evidence, not a story: give the slice's size, its error rate and the comparison with the rest.

LABORATORY

40 min

Sprint 4: run the final test once, then analyse errors

- Section 8: run once with the locked configuration and final=True
- Confirm the uuid join or the positional row count
- Make one error-analysis figure by entity, depth block or class
- Draft slides 1–4 while the numbers are fresh

NOTEBOOKS

`exercises/01_esp_gas_lock/ESP_Gas_Lock_Student.ipynb`

`exercises/02_log_curves/Log_Curve_Prediction_Student.ipynb`

`exercises/03_sensor_signals/Sensor_Signals_Student.ipynb`

EVIDENCE TO SUBMIT

- Final score table: baseline and candidate side by side
- One error-analysis figure with a caption
- Draft slides 1–4

MILESTONE 4 OF 4 · 15:25

Checkpoint 4: the locked final assessment

Show

- The final command and its configuration
- Join or alignment checks that passed
- Final baseline and candidate scores

Instructor checks

- Configuration matches checkpoint 3
- No changes after the final run
- Validation and test numbers kept apart

✓ Proceed when

- One error figure is ready
- Limitations are written down

◆ A final score below the validation score is common, not a failure: selection makes validation optimistic.

PART 7 OF 7

07

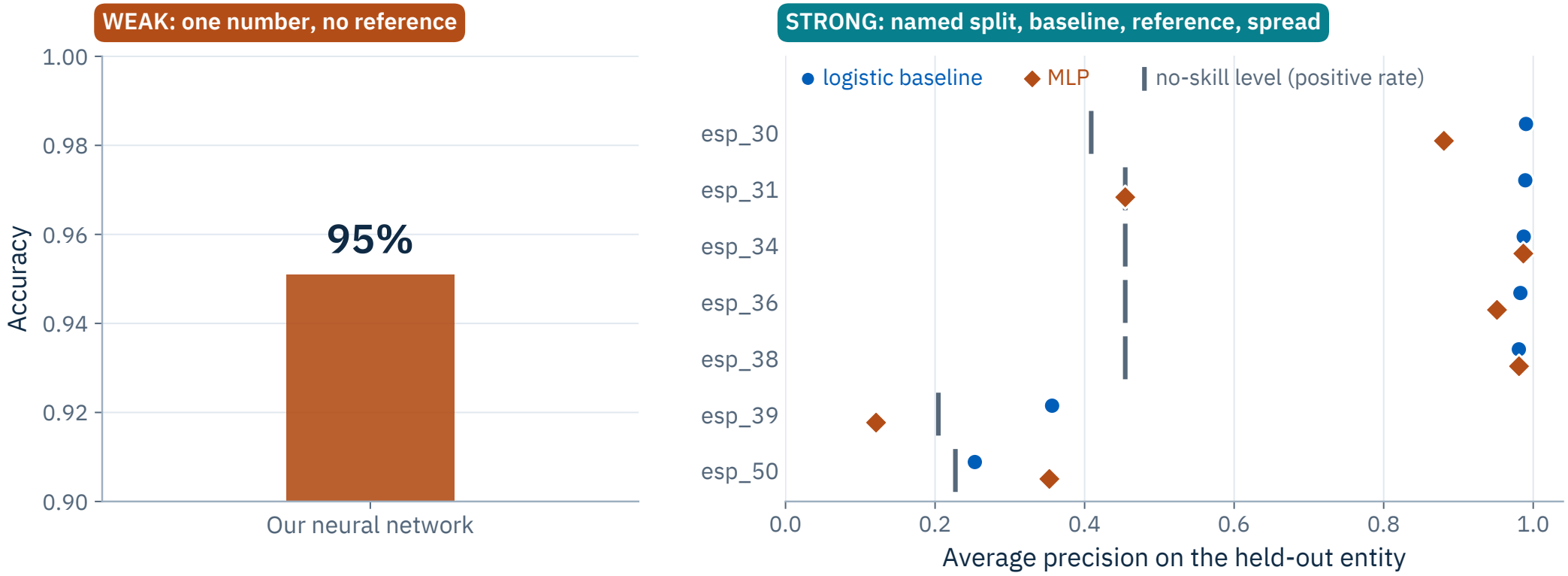
Communicating evidence

Make the claim no stronger than the evidence

- Weak and strong evidence figures
- A five-minute structure
- A model card and a reproducibility checklist
- The rubric and the handoff

Strong evidence names the split, baseline and spread

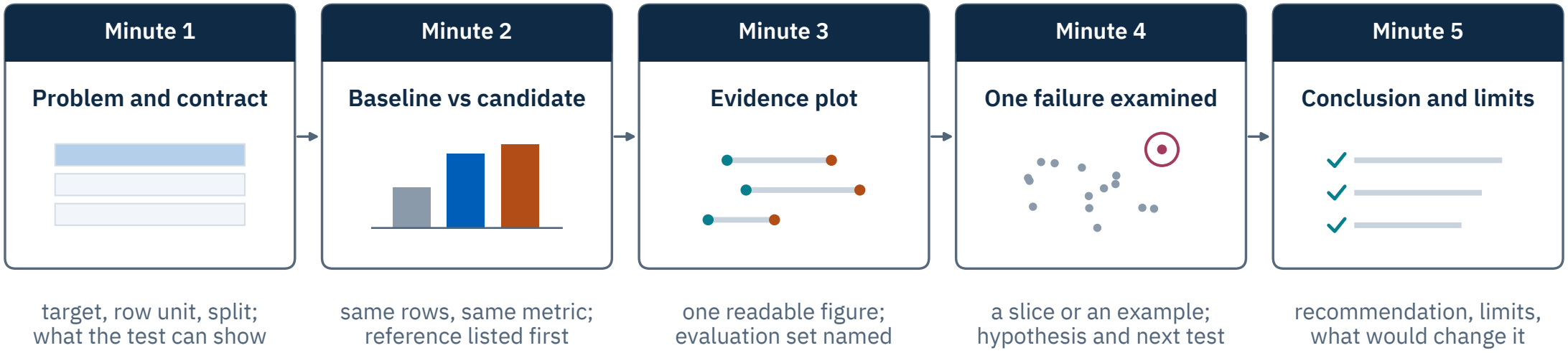
Both panels come from the same file and the same kind of network.



Computed from esp_train.csv. Left: MLP with a random 5-fold row split, pooled accuracy. Right: leave-one-entity-out average precision per entity; ticks mark each entity's positive rate.

Five minutes, five slides, one conclusion

Readable figures with axis labels, and the evaluation set named on every chart.



Schematic storyboard; two minutes of questions follow each talk.

A model card states use, evidence and limits

Model card — ESP gas-lock logistic baseline · teaching example, not a field model

Intended use

label hourly rows as gas lock or not

for entities unseen in training

Not intended for

early warning: no lead time was tested

inputs outside the training ranges

Data and split

esp_train.csv: 3,243 rows, 7 entities

leave one entity out; test entities locked

Evaluation

average precision by entity 0.25–0.99

weakest: esp_50 and esp_39

Limitations

units undocumented; 300 missing cells

labelled periods look highly regular

Reproduce

LeaveOneGroupOut on ENTITY, 7 folds

baseline pipeline of curriculum/capstone.py

Template filled with values computed from esp_train.csv (logistic baseline, leave-one-entity-out).

A reproducible run survives a fresh process

Checklist: fresh process, pinned environment, fixed seeds, saved artifacts, reload check.

fresh process

```
$ python3 curriculum/capstone.py log
    --epochs 40 --width 32
    --lr 0.001 --seed 42
    --out runs/pe_w32_s42
```

one command, pinned environment, unchanged data

Saved artifacts

- `split.csv`
- `learning_curve.csv`
- `validation_predictions.csv`
- `results.json`
- `preprocessing_and_baseline.joblib`
- `neural_weights.pt`

Checks

- ✓ reload gives identical predictions
- ✓ seed, width, lr and epochs recorded in results.json
- ✓ split ids let others rescore the run
- ! final: same flags plus --final, run once

Schematic; command-line options and artifact names from curriculum/capstone.py.

The rubric weights evaluation above accuracy

Criterion	Weight	Full-credit evidence
Problem and data contract	20%	Correct target, schema, and timing
Evaluation design	30%	Appropriate split; preprocessing fitted on training rows only
Modelling and reproducibility	20%	Baseline, candidate, saved run
Error analysis	20%	Relevant metrics and investigated failures
Communication	10%	Clear, supported conclusion and limitations

◆ Leakage or misaligned labels invalidate the affected assessment, whatever the score.

15 min

Handoff checklist: a run others can repeat

- [ ] Re-run the notebook or script from a fresh process
- [ ] Include split identifiers, preprocessing, configuration and model state
- [ ] Export predictions with identifiers or original row positions
- [ ] Submit the five-slide report and one reproducible command

EVIDENCE TO SUBMIT

- Five slides
- One command and its run folder
- Predictions with uuids or row positions

DISCUSSION

35 min

Final presentations: five minutes per team

Presenters

- Five minutes, five slides
- Evaluation set named on every chart

Audience

- One question about the evidence
- One question about a limit

Instructor

- Score with the rubric
- Note one strength and one fix

Honest negative results and clear limits are praised in front of the room.

Key takeaways

- ✓ The split decides what a score can claim; name it on every chart
- ✓ Every supplied file hid a limitation that only the audit revealed
- ✓ Baselines, including a tree ensemble, make improvement measurable
- ✓ One change over several seeds, then lock before the final test
- ✓ Errors by group, class and calibration beat one pooled number

PRACTICE

5 min · Individual

Exit check

1. A team reports 95% accuracy for ESP from a random row split. What two things do you ask for?
2. Why is a PE validation MAE not evidence of cross-well performance?
3. A network beats logistic regression by 0.003 macro-F1 on one seed. What do you conclude?
4. Name one thing you may not do after the final test run.

References and further study

- Grinsztajn, Oyallon and Varoquaux (2022). *Why do tree-based models still outperform deep learning on typical tabular data?* NeurIPS Datasets and Benchmarks Track.
- Mitchell and colleagues (2019). *Model Cards for Model Reporting.* Conference on Fairness, Accountability, and Transparency (FAT*).
- scikit-learn User Guide: cross-validation for grouped data, probability calibration, common pitfalls.
- PyTorch documentation: BCEWithLogitsLoss (pos_weight), saving and loading models.
- Course files: curriculum/capstone.py and the capstone section of INSTRUCTOR_GUIDE.md.